

Supported Zero, Heat-Flow Endpoints, and the Weil Pairing

Split-Zero research programme

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This reader calculates supported-zero inversion in the full semilattice, the escaping endpoint's exact heat coordinates, and the original arithmetic heat and Weil receivers. It retains the pole and nilpotent defects at time zero. Complete derivations, source citations and exact auxiliary checks accompany the calculations. No Riemann-hypothesis or Navier–Stokes disproof is claimed.

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We calculate three different inverse operations on the original split-zero object, retaining its complete distributive support lattice. We then construct formal heat with its exact coefficient maps, its time-zero fibre, and its support adjunctions. Two nontrivial defect objects result. First, inverting the time coordinate before taking its zero fibre leaves the full support lattice as a boundary. Second, when support is permitted independently at every coefficient, forward and reverse heat leave an explicitly calculated idempotent support closure. Neither defect is replaced by a scalar Boolean quotient. Their maps to the original amplitude heat equation are proved below.

Original sources actually used. The first source is *A Note on the Globalization Semiring* $G(\mathbb{Z})$, March 2026, original `globalization_note.tex`, whose author field is empty. The user's attribution of the originating construction to Gemini 2.5 Pro is retained separately from that byline. Its definitions, projection, and localization statements are at source lines 53–139 and 432–563; the original localization source is a prior foundation, not a new result of this paper. The semilattice and adjunction source is The Clankers, *A Secondary Note on Split-Zero Globalization: Semimodule Classification, Multiplicative Reconstruction, and All-Orders Zeta Deformation*, `split_zero_secondary_note.tex`, source lines 459–819, especially `thm:fiber-modules`, `thm:equivalence-semimodules-diagrams`, and `cor:adjunctions`. The general idempotent universal property is in The Clankers, *Split-Zero Projective Geometry and Leibniz Monads*, `split_zero_projective_monads_surcomplex.tex`, lines 358–428, `thm:idempotent-localization`. The full lattice definition and its original localization theorem are in The Clankers, full-support Version 11, Definition `def:lattice-split`, and Theorem `thm:lattice-localization`. These exact original TeX passages were read. The formal heat and coefficient-support calculations below are new derivations from those objects; they are not attributed to the source authors as prior claims.

1 The whole support lattice and the three inverses

Let R be a nonzero commutative unital ring and L a nontrivial bounded distributive lattice. Equip L with semiring addition $\vee$, multiplication $\wedge$, zero 0_L and unit 1_L . The original carrier and every one of its coefficients are

$$S_R = G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in R\} \subset R \times L. \quad (\text{SH1})$$

Write $z_\lambda = (0, \lambda)$, $\hat{a} = (a, 1_L)$, $\tau = z_{0_L}$, and $e = z_{1_L} = \hat{0}$. Operations are componentwise:

$$(a, \lambda) + (b, \mu) = (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu). \quad (\text{SH2})$$

Closure follows because an element with nonzero amplitude has support 1_L . Distributivity is precisely ring distributivity and distributivity of meet over join. The semiring zero is τ and the multiplicative unit is $\hat{1}$; both differ from e . The two projections are unital semiring homomorphisms

$$p : S_R \rightarrow R, \quad (a, \lambda) \mapsto a, \quad \chi : S_R \rightarrow L, \quad (a, \lambda) \mapsto \lambda. \quad (\text{SH3})$$

The map (p, χ) is injective with the exact image in (SH1). Thus their simultaneous receiver preserves all amplitudes and all support labels.

Proposition 1.1 (Local additive inverse). *Multiplication by e is the projector*

$$\nu : S_R \rightarrow S_R, \quad \nu(x) = ex = z_{\chi(x)}. \quad (\text{SH4})$$

The additive monoid is a join-semilattice of abelian groups: the fibre of ν at z_λ is the singleton $\{z_\lambda\}$ for $\lambda < 1_L$, and is the additive group $\{\widehat{a} : a \in R\}$ at $\lambda = 1_L$, with zero e . Its local inverse map is

$$\iota(\widehat{a}) = \widehat{-a}, \quad \iota(z_\lambda) = z_\lambda, \quad x + \iota(x) = \nu(x). \quad (\text{SH5})$$

In particular the additive inverse of e in its own fibre is e . There is no y with $e + y = \tau$.

Proof. Multiplication in (SH2) proves the projector formula. Each stated fibre is closed under addition; its indicated point is the fibre zero. Formula (SH5) proves the inverse law within every fibre. It also proves $x + \iota(x) + x = x$ and $\iota(x) + x + \iota(x) = \iota(x)$. Uniqueness of such an inverse can be checked in the two coordinates: the two equalities force the support of a candidate inverse to equal the support of x , and its amplitude must be the negative amplitude of x . Finally $\chi(e + y) = 1_L$ for every y , whereas $\chi(\tau) = 0_L$. More generally, the only invertible element of the additive monoid with respect to its global identity τ is τ itself. $\square$

Proposition 1.2 (Local multiplicative inverse). *The local multiplicative monoid at the idempotent z_λ is*

$$z_\lambda S_R z_\lambda = \{z_\mu : \mu \leq \lambda\} \cong (\downarrow \lambda, \wedge), \quad 1_{\text{local}} = z_\lambda. \quad (\text{SH6})$$

Its only unit is z_λ , with inverse z_λ . In particular the inverse of e in the local monoid $eS_R e$ is e , while the global multiplicative unit group of S_R is $\{\widehat{a} : a \in R^\times\}$.

Proof. For a supported amplitude, $z_\lambda \widehat{a} = z_\lambda$; for a zero label, $z_\lambda z_\mu = z_{\lambda \wedge \mu}$. These formulas prove the local monoid identification. If $\mu, \eta \leq \lambda$ and $\mu \wedge \eta = \lambda$, then $\mu = \eta = \lambda$, proving the unit assertion. Globally a product equal to $\widehat{1}$ must have both amplitudes nonzero and multiply to 1_R , so both factors are the claimed ring units. The equation $ey = e$ alone has all supported amplitudes as solutions; it does not say that those elements are inverses in $eS_R e$, because they do not belong to that local monoid unless they equal e . If a reflexive generalized inverse is required to satisfy $eye = e$ and $yey = y$, the first identity forces y to be supported and the second then forces $y = e$. $\square$

Theorem 1.3 (The genuine inverse-monoid structure over a field). *Let K be a field. The entire multiplicative monoid of $G_L(K)$ is an inverse monoid. Its unique generalized inverse is*

$$(\widehat{a})^\dagger = \widehat{a^{-1}} \quad (a \neq 0), \quad z_\lambda^\dagger = z_\lambda \quad (\lambda \in L). \quad (\text{SH6a})$$

It obeys $xx^\dagger x = x$ and $x^\dagger x x^\dagger = x^\dagger$ for every element, including the genuine identity $e^\dagger = e$. The corresponding multiplicative local identity is

$$\eta(x) = xx^\dagger = \begin{cases} \widehat{1}, & p(x) \neq 0, \\ z_\lambda, & x = z_\lambda. \end{cases} \quad (\text{SH6b})$$

All multiplicative idempotents form the meet-semilattice $\mathcal{E} = \{\widehat{1}\} \sqcup \{z_\lambda : \lambda \in L\}$, in which $\widehat{1}$ is above every z_λ and the original lattice L remains unchanged beneath it.

Proof. The two generalized-inverse equations follow by direct multiplication. To prove uniqueness, if $x = \widehat{a}$ with $a \neq 0$, a lower-label candidate y makes xyx have zero amplitude, and cannot give x . For $y = \widehat{b}$ the first equation gives $a^2 b = a$, hence $b = a^{-1}$ in the field. If $x = z_\lambda$ and a candidate y has nonzero amplitude, the product yxy has zero amplitude and cannot equal y . Thus $y = z_\mu$; the two equations are $\lambda \wedge \mu = \lambda$ and $\lambda \wedge \mu = \mu$, which force $\lambda = \mu$. This proves uniqueness for every element. A multiplicative idempotent with nonzero amplitude has amplitude satisfying $a^2 = a$ in a field, hence is $\widehat{1}$; all other idempotents are exactly the z_λ . The monoid is commutative, so the idempotents commute, and their multiplication is the asserted meet. This also proves the inverse-monoid assertion directly. $\square$

The map $\eta : G_L(K) \rightarrow \mathcal{E}$ is a multiplicative monoid homomorphism: a product of nonzero field amplitudes is nonzero, and every remaining product meets its zero labels. The full original support map factors through it by $\widehat{1} \mapsto 1_L$, $z_\lambda \mapsto \lambda$. This last map identifies the two different idempotents $\widehat{1}$ and e ; η itself retains their difference. At an actual supported scalar difference $d = \widehat{a - b}$ one has the exact two cases

$$dd^\dagger = \begin{cases} \widehat{1}, & a \neq b, \\ e, & a = b. \end{cases} \quad (\text{SH6c})$$

Thus inverse-monoid division by the supported zero is well-defined, with its specified local identity e . For any $f \in G_L(K)$ the exact division equation is $d(d^\dagger f) = \eta(d)f$. On the distinct-value locus it is f ; at $a = b$ it is $ef = \nu(f)$. This supplies the precise idempotent defect in a rational identity whose ordinary derivation canceled d . It does not assume that the unit $\widehat{1}$ remains its local identity at a collision.

Theorem 1.4 (Universal unital inversion). *For every $\lambda \in L$, the universal unital localization is*

$$S_R[z_\lambda^{-1}] \cong \downarrow \lambda, \quad q_\lambda(x) = \lambda \wedge \chi(x). \quad (\text{SH7})$$

The target has unit λ . In particular

$$S_R[e^{-1}] \cong L, \quad q_1 = \chi, \quad S_R[\tau^{-1}] \cong \{0\}. \quad (\text{SH8})$$

The localization at e keeps every label of L and identifies all supported amplitudes with 1_L .

Proof. For any commutative semiring S and idempotent a , the subset aS is a semiring with unit a and the same zero as S . The map $x \mapsto ax$ is unital into this new target because $\widehat{1}$ maps to its unit a . If $f : S \rightarrow T$ is unital and $f(a)$ is invertible, $f(a)^2 = f(a)$ forces $f(a) = 1_T$. It follows that $ax = ay$ implies $f(x) = f(y)$, so $\bar{f}(ax) = f(x)$ is a well-defined unique factorization. This proves the universal property of aS . Apply it to $a = z_\lambda$ and use (SH6). The two endpoint cases are $\lambda = 1_L, 0_L$. $\square$

The kernel *congruence* of χ is exactly

$$\mathcal{R}_\chi = \{(\widehat{a}, \widehat{b}) : a, b \in R\} \cup \{(z_\lambda, z_\lambda) : \lambda < 1_L\}. \quad (\text{SH9})$$

Its zero ideal $\chi^{-1}(0_L) = \{\tau\}$ does not detect the identified amplitudes. The kernel-pair object $S_R \times_L S_R$, its two projections to S_R , and the amplitude difference

$$d : S_R \times_L S_R \rightarrow (R, +), \quad d(x, y) = p(x) - p(y)$$

give an exact description of that identification: d is additive and surjective, and $d^{-1}(0)$ is the diagonal copy of S_R . Surjectivity uses $(\widehat{a}, e)$; the kernel assertion follows from injectivity of (p, χ) . This supplies an explicit receiver for the amplitudes identified by localization.

The ring reflection p is universal among unital maps to rings. Indeed $e + z_\lambda = e$ forces every z_λ to map to zero in a ring by additive cancellation; the remaining restriction is then a ring homomorphism. Consequently the pushout of $S_R \rightarrow L$ and $S_R \rightarrow R$ in commutative unital semirings is the one-element ring: in the pushout the same element e is both 1 and 0. This proves the exact relation between the support localization and the ring reflection.

2 The original semilattice adjunctions with the full lattice

Let M be a unital S_R -semimodule, meaning a commutative monoid under addition with a distributive S_R action. Its global zero is 0_M . Put

$$\epsilon_M(m) = em, \quad P_M = eM, \quad M_\ell = \{m : em = \ell\} \quad (\ell \in P_M). \quad (\text{SH10})$$

The module support semilattice P_M is distinguished from the base support lattice L .

Theorem 2.1 (Full-lattice semilattice description). *The set P_M is a join-semilattice with zero 0_M under the addition in M , and each M_ℓ is an R -module with zero ℓ . For $\ell \leq \ell'$ the transition*

$$T_{\ell, \ell'} : M_\ell \rightarrow M_{\ell'}, \quad m \mapsto m + \ell' \quad (\text{SH11})$$

is R -linear and these transitions compose. The remaining full lattice action is precisely an L -semimodule action on P_M :

$$\lambda \cdot \ell = z_\lambda \ell, \quad z_\lambda m = z_\lambda(em) = \lambda \cdot \ell \in P_M \quad (m \in M_\ell). \quad (\text{SH12})$$

Conversely, a join-semilattice P with zero carrying an L -semimodule action, together with a diagram of R -modules M_ℓ and R -linear transitions over P , reconstructs an S_R -semimodule by the following explicit operations:

$$\begin{aligned} M &= \bigsqcup_{\ell \in P} M_\ell, & m + n &= T_{\ell, \ell \vee \eta} m + T_{\eta, \ell \vee \eta} n \quad (m \in M_\ell, n \in M_\eta), \\ \widehat{a} m &= am \in M_\ell, & z_\lambda m &= 0_{M_{\lambda \cdot \ell}}. \end{aligned} \quad (\text{SH13})$$

The overlap $z_{1_L} = \widehat{0}$ has identical actions in these two formulas.

Proof. The identities $e^2 = e$, $e + e = e$, and $\widehat{1} + e = \widehat{1}$ give $\epsilon_M^2 = \epsilon_M$, $em + em = em$, and $m + em = m$. Thus P_M is the asserted semilattice. In a fibre M_ℓ , the last identity makes ℓ its zero, and $m + \widehat{1}m = em = \ell$ proves the additive inverse. Supported scalars preserve the fibre because $e\widehat{a} = e$, giving its R -module structure. The formula for transitions preserves fibres, zero, addition and scalar multiplication; composing transitions uses $\ell + \ell' = \ell'$ for $\ell \leq \ell'$. Addition between fibres is their sum transported to the join fibre because $em + en$ is that join. These are the original semilattice constructions.

For the full L action, $z_\lambda e = z_\lambda$ proves (SH12). All semimodule laws on P_M follow from the operations on the zero fibre of S_R . In particular $\lambda \cdot \ell \leq \ell$, since $\lambda \vee 1_L = 1_L$. Conversely the usual diagram sum in (SH13) is associative: transport three summands to their common join, where ordinary module associativity applies. Zero, scalar multiplication by supported amplitudes, and fibre inverses follow from their module counterparts. Scalar identities within L are the L -semimodule identities on P . For mixed scalars, $(\widehat{a} + z_\lambda)m = \widehat{a}m$ holds because the zero in fibre $\lambda \cdot \ell \leq \ell$ transports to the zero in M_ℓ . The identity $\widehat{a}z_\lambda = z_\lambda$ holds because an R scalar sends every fibre zero to itself. The zero scalar z_{0_L} acts as the zero of M_{0_P} . Thus every semimodule law holds. Extraction and reconstruction return the same elements and all the same operations. $\square$

The adjunctions retain all these support fibres:

$$\begin{aligned} \text{Hom}_{S_R}(M, \chi^* N) &\cong \text{Hom}_L(eM, N), \\ K_M &= \{m : em = 0_M\} = M_{0_M}, & \text{Hom}_{S_R}(p^* A, M) &\cong \text{Hom}_R(A, K_M). \end{aligned} \quad (\text{SH14})$$

Here N is an L -semimodule and A an R -module. For the first identity, $f(m) = ef(m) = f(em)$ forces unique factorization through $m \mapsto em$; conversely an L -linear map on eM composed with this projector is S_R -linear. For the second, $eg(a) = g(ea) = g(0) = 0_M$ forces the image into K_M ; on that fibre every z_λ acts by zero since $z_\lambda m = z_\lambda em$, so precisely the R -linear maps occur. These arguments prove both bijections and their naturality.

For the regular module $M = S_R$, one has $P_M = L$, top fibre R , and singleton fibres below the top. In particular K_{S_R} is the singleton $\{\tau\}$, whereas the ring reflection is R . The right adjoint K_M is not being substituted for the amplitude projection p . The first adjoint is precisely universal inversion of the idempotent action: if an S_R -linear map receives e as an invertible operator, that idempotent operator must be the identity, so the map factors uniquely through eM .

3 The exact integral coefficient heat operator

Let $A = R[[t, z]]$, with t the formal heat parameter. An element F is a family of coefficients $f_{mn} \in R$: $F = \sum_{m,n \geq 0} f_{mn} t^m z^n$. The variable z is the unchanged spatial coordinate. For $\sigma \in \{+1, -1\}$ define

$$[t^m z^n] \mathcal{H}_\sigma F = \sum_{j=0}^m \sigma^j \frac{(n+2j)!}{n! j!} f_{m-j, n+2j}. \quad (\text{SH15})$$

Each displayed integer is integral: $(n+2j)!/(n!j!) = \binom{n+2j}{2j} (2j)!/j!$. Thus the definition makes sense over every R , with no division operation in R . Every coefficient is a finite sum. The notation $\mathcal{H}_\sigma = \exp(\sigma t \partial_z^2)$ records this definition, not an analytic convergence assertion. The Newman heat sign $e^{tu^2} \cos(zu)$ corresponds to $\sigma = -1$, because $\partial_z^2 \cos(zu) = -u^2 \cos(zu)$.

Theorem 3.1 (Formal inverse and time-zero value). *The maps $\mathcal{H}_\sigma$ are $R[[t]]$ -linear continuous automorphisms for the t -adic topology, and*

$$\mathcal{H}_+ \mathcal{H}_- = \mathcal{H}_- \mathcal{H}_+ = I, \quad \text{ev}_0 \mathcal{H}_\sigma = \text{ev}_0, \quad \text{ev}_0(F) = F(0, z). \quad (\text{SH16})$$

They commute with ∂_z and with every coefficient-ring homomorphism. For initial data $f(z) \in R[[z]]$, $F = \mathcal{H}_\sigma f$ satisfies

$$\partial_t F = \sigma \partial_z^2 F, \quad F(0, z) = f(z). \quad (\text{SH17})$$

When R is a $\mathbb{Q}$ -algebra these conditions uniquely determine F . All assertions extend to coefficients in any R -module.

Proof. The formula uses only integer scalar maps, sums and shifts of indices, proving linearity, coefficient naturality and commutation with ∂_z . It preserves the ideal $t^k A$ for every k ; and multiplication by t commutes with it, proving continuity and $R[[t]]$ -linearity. To calculate the inverse, collect terms whose two heat indices sum to k . Over $\mathbb{Q}$ their coefficient in the composite is

$$\frac{(n+2k)!}{n!} \sum_{j=0}^k \frac{(-1)^j}{j!(k-j)!} = \frac{(n+2k)!}{n!k!} (1-1)^k.$$

Every summand before collection is an integer, being a product of the integer coefficients in (SH15); therefore this integer identity holds over every R . It is 1 at $k=0$ and zero otherwise, proving both inverse formulas. At $m=0$ only $j=0$ occurs, proving evaluation at zero.

For initial data the coefficients are $[t^m z^n] F = \sigma^m (n+2m)!/(n!m!) f_{n+2m}$. Multiplying the coefficient with index $m+1$ by $m+1$ gives exactly $\sigma(n+2)(n+1)$ times the coefficient with indices $m, n+2$. This proves the differential equation without division in R . If all positive integers are invertible in R , the recurrence $(m+1)F_{m+1}(z) = \sigma F_m''(z)$ uniquely constructs the solution from $F_0 = f$. Module coefficients obey the same identities, proving the final assertion. $\square$

The $\mathbb{Q}$ -algebra condition is needed for that PDE uniqueness claim, not for (SH15) or its inverse. For example over $R = \mathbf{F}_p$, the nonzero series t^p has zero time-zero value and satisfies $\partial_t t^p = \sigma \partial_z^2 t^p = 0$. The exact additional solution object in any characteristic is

$$\mathcal{N}_R = \{(F_m)_{m \geq 0} : F_0 = 0, (m+1)F_{m+1} = \sigma F_m''\} \subset tR[[t, z]]. \quad (\text{SH18})$$

The difference of any two solutions with the same initial value lies in $\mathcal{N}_R$, and adding an element of $\mathcal{N}_R$ to a solution gives another with that initial value. The $\mathbb{Q}$ -algebra calculation proves $\mathcal{N}_R = 0$ there.

The heat map is generally not a ring homomorphism. With the original coordinate z and sign σ ,

$$\mathcal{H}_\sigma z = z, \quad \mathcal{H}_\sigma(z^2) = z^2 + 2\sigma t, \quad \mathcal{H}_\sigma(z^2) - (\mathcal{H}_\sigma z)^2 = 2\sigma t. \quad (\text{SH19})$$

Its complete multiplication defect has an exact receiver:

$$\mathcal{H}_\sigma(FG) - (\mathcal{H}_\sigma F)(\mathcal{H}_\sigma G) = \sum_{k \geq 1} \frac{(2\sigma t)^k}{k!} \partial_z^k(\mathcal{H}_\sigma F) \partial_z^k(\mathcal{H}_\sigma G). \quad (\text{SH20})$$

This is coefficientwise integral as well: the product of the two derivative coefficients is divisible by $k!$ as an integer. To prove the identity, first work over $\mathbb{Q}$ with independent variables z_1, z_2 , so that differentiation of the product is $\partial_1 + \partial_2$. The three commuting differential operators in $(\partial_1 + \partial_2)^2 = \partial_1^2 + 2\partial_1\partial_2 + \partial_2^2$ exponentiate and give the formula, including its $k = 0$ term. Each fixed output coefficient involves finitely many input coefficients and has integral coefficients on both sides; the identity in their polynomial ring over $\mathbb{Q}$ is consequently an integral identity, so descends to every R .

An exact repaired multiplication, rather than an imposed multiplicativity claim, is

$$F \star_\sigma G = \sum_{k \geq 0} \frac{(2\sigma t)^k}{k!} (\partial_z^k F)(\partial_z^k G) = \mathcal{H}_\sigma((\mathcal{H}_{-\sigma} F)(\mathcal{H}_{-\sigma} G)). \quad (\text{SH21})$$

Transport of multiplication through the proved additive automorphism makes this an associative commutative $R[[t]]$ -algebra with unit 1, and makes $\mathcal{H}_\sigma : (A, \cdot) \rightarrow (A, \star_\sigma)$ a unital ring isomorphism. At $t = 0$ this multiplication is the original one. This construction retains the precise product defect and does not assume that the original heat map was multiplicative.

Multiplicative reciprocals of actual units have an equally exact map. A series $F \in A$ is a unit precisely when its coefficient f_{00} is a unit in R : necessity follows by constant-term evaluation, and sufficiency follows from the coefficientwise convergent geometric series after factoring out f_{00} . Heat preserves this constant coefficient, hence preserves the set of units. For every such F ,

$$\begin{aligned} (\mathcal{H}_\sigma F) \star_\sigma \mathcal{H}_\sigma(F^{-1}) &= 1, \\ (\mathcal{H}_\sigma F) \mathcal{H}_\sigma(F^{-1}) - 1 &= - \sum_{k \geq 1} \frac{(2\sigma t)^k}{k!} \partial_z^k(\mathcal{H}_\sigma F) \partial_z^k(\mathcal{H}_\sigma(F^{-1})). \end{aligned} \quad (\text{SH21a})$$

The first is the transported multiplication law; the second is (SH20) applied to $FF^{-1} = 1$. This proves the complete ordinary-inversion defect and its divisibility by t , so at time zero the ordinary reciprocal is retained. For example over $\mathbb{Q}$, $F = 1 + z$ has $\mathcal{H}_\sigma F = 1 + z$, whereas $\mathcal{H}_\sigma(F^{-1}) = (1 + z)^{-1} + 2\sigma t(1 + z)^{-3} + O(t^2)$. The defect is already nonzero at the displayed first order.

The field inverse-monoid theorem is applied in its stated coefficient category. The formal series ring $K[[t, z]]$ is not a field: t has no reflexive multiplicative inverse in it, since $tBt = t$ would imply $tB = 1$ in this integral domain, contradicting evaluation at $t = 0$. The exact extension receiving its inverse is the punctured chart in (SH28), whose full supported time-zero fibre is proved below.

4 Heat on the full semilattice and its adjunction maps

On the actual requested carrier $G_L(A)$ define

$$\widetilde{\mathcal{H}}_\sigma(\widehat{F}) = \widehat{\mathcal{H}_\sigma F}, \quad \widetilde{\mathcal{H}}_\sigma(z_\lambda) = z_\lambda. \quad (\text{SH22})$$

The two prescriptions agree at $e = \widehat{0} = z_{1_L}$. They give a $G_L(R[[t]])$ -linear automorphism of the underlying semimodule. Indeed additivity follows from heat additivity on the top fibre and from preservation of all lower zero fibres; supported scalars act by $R[[t]]$ multiplication, while z_λ sends every top-fibre element to z_λ and meets lower labels. Both operations commute with (SH22).

Its inverse is $\tilde{\mathcal{H}}_{-\sigma}$. It is generally not a homomorphism for the original semiring product, by (SH19); it is a unital semiring isomorphism to $G_L(A, \star_\sigma)$ by (SH21).

The exact receiving diagram is

$$\begin{array}{ccccc} A & \xleftarrow{p} & G_L(A) & \xrightarrow{\chi} & L \\ \mathcal{H}_\sigma \downarrow & & \tilde{\mathcal{H}}_\sigma \downarrow & & \downarrow I_L \\ A & \xleftarrow{p} & G_L(A) & \xrightarrow{\chi} & L. \end{array} \quad (\text{SH23})$$

Both squares commute directly from the coefficient formulas. Here the vertical maps are additive heat maps; the right square is also the exact induced map after universal unital inversion of e . On that localization the induced map is I_L . The restriction to the local monoid $eG_L(A)e = L$ is likewise I_L , and heat commutes with every local additive inverse in (SH5). For a general label the localized square is the same with L replaced by $\downarrow \lambda$ and χ replaced by q_λ . No Boolean branch has been chosen.

The construction also works on the original general semilattice diagram, without replacing it by the regular scalar fibre. For the data in Theorem 2.1, replace each module M_ℓ by $M_\ell[[t, z]]$ and each transition by its coefficientwise extension. The lattice P and its complete L action are unchanged. The heat maps of Theorem 3.1 commute with all these transitions because their coefficients are integers. They thus form a natural automorphism of the reconstructed semimodule. Under the two adjunctions (SH14), they induce the identity on the support object P and the ordinary coefficient heat on $M_{0P}[[t, z]]$, respectively. These claims follow by applying the explicit factorization maps in the proof of (SH14), not by replacing one adjoint with the other.

The derivative lifts also preserve every support label. Hence the differential equation is an equality in each module fibre: for example on the top fibre $\tilde{\partial}_t \hat{F} = \sigma \tilde{\partial}_z^2 \hat{F}$ means exactly $\partial_t F = \sigma \partial_z^2 F$. Bringing its two sides together gives the fibre zero e , not the external zero τ . On a lower singleton fibre both sides equal z_λ . This is the same fibrewise inverse operation as in (SH5).

5 The exact time-zero fibre and its punctured-time boundary

Set $A_0 = R[[z]]$. The time-zero map is the unital semiring map

$$\tilde{\text{ev}}_0 : G_L(A) \rightarrow G_L(A_0), \quad \hat{F} \mapsto \widehat{F(0, z)}, \quad z_\lambda \mapsto z_\lambda. \quad (\text{SH24})$$

It is onto, with constant-in- t section. Its kernel congruence has the following complete description:

$$\hat{F} \sim_0 \hat{G} \iff F - G \in tA, \quad z_\lambda \sim_0 z_\mu \iff \lambda = \mu; \quad (\text{SH25})$$

there is no additional identification between a lower label and a supported amplitude. In particular

$$\tilde{\text{ev}}_0^{-1}(\tau) = \{\tau\}, \quad \tilde{\text{ev}}_0^{-1}(e) = \{\widehat{tF} : F \in A\}. \quad (\text{SH26})$$

Thus the semiring-zero kernel ideal is trivial, while its congruence retains the full amplitude ideal tA . The ordinary exact sequence is $0 \rightarrow tA \rightarrow A \rightarrow A_0 \rightarrow 0$, and its full supported lift is precisely (SH25), not a quotient obtained by identifying supported zeros with τ .

The complete time-zero heat calculation is

$$\begin{array}{ccc} G_L(A) & \xrightarrow{\tilde{\text{ev}}_0} & G_L(A_0) \\ \tilde{\text{ev}}_0 \tilde{\mathcal{H}}_\sigma = \tilde{\text{ev}}_0, & \chi \downarrow & \downarrow \chi \\ L & \xrightarrow{I_L} & L. \end{array} \quad (\text{SH27})$$

Both claims follow from the $m = 0$ term in (SH15) and preservation of all labels. Thus localization at e and evaluation at $t = 0$ commute through the displayed full-lattice map.

Theorem 5.1 (Supported special fibre of punctured time). *Let $\hat{t} = (t, 1_L) \in G_L(A)$. There are canonical isomorphisms of unital semirings*

$$G_L(A)[\hat{t}^{-1}] \cong G_L(A[t^{-1}]), \quad G_L(A)[\hat{t}^{-1}] \amalg_{G_L(A)} G_L(A_0) \cong L. \quad (\text{SH28})$$

The second target is the actual pushout for time-zero base change of the punctured-time chart. For every $n \geq 0$,

$$G_L(A/(t^{n+1}))[\hat{t}^{-1}] \cong L. \quad (\text{SH29})$$

All transition maps between these localized finite thickenings are the identity on L .

Proof. Use fractions with denominators $\hat{t}^k$. The map $z_\lambda/\hat{t}^k \mapsto z_\lambda$ and $\hat{F}/\hat{t}^k \mapsto \widehat{F/\hat{t}^k}$ is surjective and respects both operations. Two fractions with lower labels are equal exactly when their labels agree, since multiplication by a supported denominator leaves such a label unchanged. A lower-label fraction cannot equal a top fraction. For two top fractions, their equality is exactly the usual ring localization relation. This proves the first isomorphism.

For any unital target T , compatible maps from the first pushout factors are exactly maps $G_L(A_0) \rightarrow T$ in which the image of $\hat{t}$ under time-zero evaluation is invertible. That image is e , so this universal property is the property of $G_L(A_0)[e^{-1}]$. Theorem 1.4 identifies it with L , proving the second isomorphism. In the n th thickening, $\hat{t}^{n+1} = e$, so inverting $\hat{t}$ forces e to be invertible. Conversely, after applying χ every supported amplitude, including $\hat{t}$, is already the unit 1_L . These two universal factorizations prove (SH29); they also force the stated identity transition maps. $\square$

The maps to the pushout in (SH28) are $\widehat{F/\hat{t}^k} \mapsto 1_L$ and $z_\lambda \mapsto \lambda$ from the punctured chart, and $\hat{f} \mapsto 1_L$, $z_\lambda \mapsto \lambda$ from its time-zero chart. Thus every amplitude is explicitly identified there, but no label is. On ordinary ring amplitudes the corresponding pushout is the zero ring, since an invertible t and $t = 0$ force $1 = 0$. This proves the exact map from the retained support boundary to the ordinary empty ring fibre. It is not a statement that the supported boundary is absent.

The spectral map retains its exact geometry. Every ideal of $G_L(A)$ either lies in the zero-label part, where it is a lattice ideal, or contains a supported amplitude and hence contains e and all z_λ ; its amplitudes then form a ring ideal. Testing products gives the two prime families $P_{\mathfrak{p}} = \{z_\lambda : \lambda \in \mathfrak{p}\}$ and $Q_{\mathfrak{q}} = \{z_\lambda : \lambda \in L\} \cup \{\hat{F} : F \in \mathfrak{q}\}$. Every $P_{\mathfrak{p}}$ omits e , and every $Q_{\mathfrak{q}}$ contains it. The basic-open formulas $D(z_\lambda) = D_L(\lambda)$ and $D(\hat{F}) = \text{Spec}_{\text{lat}} L \cup D_A(F)$ prove

$$\text{Spec } G_L(A) = \text{Spec}_{\text{lat}} L \blacktriangleright \text{Spec } A, \quad D(e) = \text{Spec}_{\text{lat}} L. \quad (\text{SH30})$$

The map on spectra from the pushout fibre L in (SH28) is exactly this full support stratum in the time-zero chart. It does not select one Boolean character.

The time-zero congruence map itself has the precise spectral image

$$\text{im}(\text{Spec } G_L(A_0) \rightarrow \text{Spec } G_L(A)) = \text{Spec}_{\text{lat}} L \cup V_A(t), \quad D(\hat{t}) = \text{Spec}_{\text{lat}} L \cup D_A(t). \quad (\text{SH30a})$$

Indeed pure-support primes pull back to the same lattice primes, and an arithmetic prime pulls back along $A \rightarrow A/(t)$ to a prime containing t . These are all such primes by the ordinary quotient correspondence. Their intersection with the punctured chart is exactly $\text{Spec}_{\text{lat}} L$. Thus the categorical fibre in (SH28) also has this explicit topological map. The time-zero quotient is the stated congruence quotient; it is not being identified with the semiring ideal quotient by its trivial zero ideal in (SH26).

Heat also extends to the punctured chart as an additive map: $\mathcal{H}_\sigma(F/t^k) = \mathcal{H}_\sigma(F)/t^k$. $R[[t]]$ -linearity makes the definition independent of the chosen denominator; its inverse is the opposite heat sign. Its support projection to the boundary L is the identity. All time thickenings before localization still keep their amplitudes: heat preserves $t^{n+1}A$ and induces inverse maps on $A/(t^{n+1})$. The inverse limit of their full split carriers is $G_L(A)$, because a compatible sequence has one fixed support label, and at top support is a compatible sequence of ring coefficients. This describes both the retained formal neighborhood and its localized support-only boundary.

6 Independent coefficient supports and the exact heat defect

The requested global carrier $G_L(R[[t, z]])$ has one support label for an entire series. The original semilattice theory also allows the larger coefficient-series semiring

$$\mathcal{T} = (G_L(R))[[t, z]].$$

An element X has at each bidegree a pair $X_{mn} = (f_{mn}, \lambda_{mn}) \in G_L(R)$. Addition is coefficientwise, and multiplication is the finite Cauchy sum. Its amplitude map goes to $R[[t, z]]$ and its coefficient support map goes to the full semiring $L[[t, z]]$. Neither collection of labels is collapsed.

Define coefficient heat using the same integers as before, lifted as supported scalars:

$$(\mathcal{H}_\sigma X)_{mn} = \sum_{j=0}^m \sigma^j (n + 2j)! / (n! j!) X_{m-j, n+2j}. \quad (\text{SH31})$$

Even when that integer is zero in R , its displayed lift is e , not τ . Thus (SH31) has the exact amplitude and support formulas

$$p(\mathcal{H}_\sigma X) = \mathcal{H}_\sigma p(X), \quad \chi(\mathcal{H}_\sigma X)_{mn} = (C\lambda)_{mn} := \bigvee_{j=0}^m \lambda_{m-j, n+2j}. \quad (\text{SH32})$$

These are finite joins in L , so no completeness hypothesis on the lattice is needed.

Theorem 6.1 (The complete coefficient-support closure). *Define $\mathcal{C} : \mathcal{T} \rightarrow \mathcal{T}$ by keeping the amplitude of each coefficient and replacing its label by $C\lambda$ in (SH32). Then*

$$\mathcal{H}_+ \mathcal{H}_- = \mathcal{H}_- \mathcal{H}_+ = \mathcal{C}, \quad \mathcal{C}^2 = \mathcal{C}, \quad \mathcal{C} \mathcal{H}_\sigma = \mathcal{H}_\sigma \mathcal{C} = \mathcal{H}_\sigma. \quad (\text{SH33})$$

All these maps are additive $G_L(R)$ -linear maps. The image and fixed set of $\mathcal{C}$ coincide, and are specified exactly by

$$\lambda_{m-1, n+2} \leq \lambda_{mn} \quad (m \geq 1, n \geq 0). \quad (\text{SH34})$$

On that retained-support subspace the two heat signs are inverse automorphisms. On the entire coefficient space their failure to be inverse is exactly $\mathcal{C}$, with no amplitude defect.

Proof. The replacement gives a valid coefficient of $G_L(R)$ because the join includes its original label; any nonzero original amplitude already has label 1_L . The amplitude of either composite heat is the original one by (SH16). Its support is

$$\bigvee_{i=0}^m \bigvee_{j=0}^{m-i} \lambda_{m-i-j, n+2i+2j} = \bigvee_{k=0}^m \lambda_{m-k, n+2k},$$

because every $k = i + j$ from 0 to m occurs. This proves the first identity and the idempotence of the support closure. Applying this calculation to one heat image proves the two commutation identities. Distributivity of finite joins over joins proves additivity on support; amplitudes are additive. For a scalar with label η , distributivity $\eta \wedge \bigvee_j \lambda_j = \bigvee_j (\eta \wedge \lambda_j)$ proves scalar compatibility, including supported scalar zero.

The fixed condition implies (SH34) because the defining join includes $j = 1$. Conversely that inequality iterated j times gives $\lambda_{m-j, n+2j} \leq \lambda_{mn}$ for every j , so the join equals the term with $j = 0$. This proves the exact fixed set. Idempotence identifies it with the image; the first identity in (SH33) then proves mutual inverse heat maps on it. $\square$

The exact quotient object of this defect is also explicit. The two heat maps and the retraction $\mathcal{C}$ have the same kernel equivalence:

$$X \sim_{\mathcal{C}} Y \iff p(X) = p(Y) \text{ and } \mathcal{C}\chi(X) = \mathcal{C}\chi(Y). \quad (\text{SH34a})$$

For $\mathcal{C}$ this follows from its coordinate definition. For heat, equality of its images gives equality of input amplitudes by the inverse amplitude map, and equality of closed supports by (SH32); the converse is the same formula. Thus the quotient additive $G_L(R)$ -semimodule $\mathcal{T}/\sim_{\mathcal{C}}$ is isomorphic to $\text{im } \mathcal{C}$, with its two induced inverse heat maps. Inclusion and $\mathcal{C}$ give a specified split retraction onto that image. The preimage of the external zero under each heat map remains the single zero series, since every original label occurs in its closure. Thus its possible noninjectivity is encoded by (SH34a), not by its zero ideal.

For a literal nonzero defect, take the coefficient monomial $X = z^2$ in $\mathcal{T}$, meaning coefficient $\hat{1}$ at $(m, n) = (0, 2)$ and τ at every other bidegree. Then

$$\mathcal{C}X : \quad X_{0,2} = \hat{1}, \quad X_{1,0} = e, \quad X_{mn} = \tau \text{ at every other bidegree.} \quad (\text{SH35})$$

The amplitude is still z^2 , but the new coefficient e at t is a retained supported zero. This follows directly from (SH32): the only additional path from $(0, 2)$ is to $(1, 0)$. Over characteristic zero it is also the explicit cancellation of the $2t$ and $-2t$ amplitudes in the two heat steps. The same support computation remains valid in characteristic two, where each of those amplitudes is already zero but its coefficient is still supported.

There is also an exact infinite even-series example over $\mathbb{Q}$. Let X^{init} have coefficient $(-1)^n/(2n)!$ at $(0, 2n)$ for every $n \geq 0$, and τ elsewhere. Its amplitude is $\cos z$ and its initial time support is exactly the zeroth row. Then

$$p(\mathcal{H}_\sigma X^{\text{init}}) = e^{-\sigma t} \cos z, \quad [t^m z^{2n}] p(\mathcal{H}_\sigma X^{\text{init}}) = \frac{(-\sigma)^m (-1)^n}{m! (2n)!} \neq 0.$$

This follows either by direct substitution in (SH15), using $\partial_z^{2m} \cos z = (-1)^m \cos z$, or by the displayed coefficient formula itself. Its support is 1_L at every $(m, 2n)$ and 0_L at every odd spatial index. That support is $\mathcal{C}$ -fixed by (SH34). Reversing the heat therefore gives the initial amplitude $\cos z$ with supported zero e at every positive-time even coefficient. It is exactly $\mathcal{C}X^{\text{init}}$, not X^{init} . This explicitly realizes the even-coefficient support cone and its retained zero coefficients.

There is an exact comparison map from the global carrier:

$$\Gamma : G_L(R[[t, z]]) \hookrightarrow \mathcal{T}, \quad \Gamma(F, \lambda)_{mn} = (f_{mn}, \lambda). \quad (\text{SH36})$$

For $\lambda < 1_L$, necessarily every $f_{mn} = 0$. The map is injective, additive and multiplicative: at bidegree (m, n) every Cauchy summand has the same meet of the two global labels, and there is a nonempty finite set of summands. It need not send the unit to the standard unit of $\mathcal{T}$. Its image unit is the idempotent E with coefficient $\hat{1}$ at $(0, 0)$ and e everywhere else. Thus Γ is a unital semiring map to its image, with that explicit unit, rather than a unital map to the whole standard power-series semiring. The image is the exact subsemiring of series with one constant label at every coefficient. It is fixed by $\mathcal{C}$, and

$$\mathcal{H}_\sigma \Gamma = \Gamma \tilde{\mathcal{H}}_\sigma, \quad \mathcal{C} \Gamma = \Gamma. \quad (\text{SH37})$$

The amplitude identity is (SH15); the support identity is the join of finitely many identical labels. In particular the forward-reverse defect on arbitrary coefficient supports does not contradict the actual automorphism on the specified global-label carrier.

The coefficientwise support quotient itself is the universal inversion of the constant supported scalar e in $\mathcal{T}$:

$$\mathcal{T}[e^{-1}] \cong L[[t, z]], \quad X \mapsto \sum_{m,n} \lambda_{mn} t^m z^n. \quad (\text{SH38})$$

Indeed multiplication by that constant e sends every coefficient to its zero label and leaves its support unchanged; its image is exactly the displayed semiring, with the usual constant unit. The idempotent localization proof applies. On this quotient either heat sign induces the same map $\mathcal{C}$, not an inverse pair. This is the complete support receiver of the coefficient defect. Here the constant scalar being inverted has coefficient e at $(0,0)$ and τ at every other index; it differs from $\Gamma(e)$, whose every coefficient is e . Their connecting map is explicit: the support projection of Γ is $\Lambda : L \rightarrow L[[t, z]]$, with $\Lambda(\lambda)_{mn} = \lambda$ at every index. This map preserves both operations and has unit the all- 1_L series in its image, rather than the standard constant unit. It satisfies $\mathcal{C}\Lambda = \Lambda$ and $\chi_{\text{coeff}}\Gamma = \Lambda\chi$. Thus the two universal support receivers are connected without identifying their different coefficient units.

Finally time-zero evaluation in $\mathcal{S}$ keeps exactly the row $m = 0$. That row of (SH32) has only $j = 0$, so

$$\text{ev}_0 \mathcal{H}_\sigma = \text{ev}_0, \quad \text{ev}_0 \mathcal{C} = \text{ev}_0, \quad \text{ev}_0 \Gamma = \Gamma_0 \widetilde{\text{ev}}_0. \quad (\text{SH39})$$

Here $\Gamma_0 : G_L(R[[z]]) \rightarrow G_L(R)[[z]]$ is the same constant-label coefficient map in one variable. Thus even the enlarged support defect has the original time-zero coefficient fibre, with every original label retained.

7 The exact receiver of zero locations

Formal substitution $z = a$ into an arbitrary element of $R[[z]]$ is not assumed. For an analytic initial function $h(z)$ and a specified complex point a , instead use its actual local coordinate $w = z - a$ and Taylor series $h(a + w) \in \mathbb{C}[[w]]$. This yields a unital evaluation map on the formal germ:

$$G_L(\mathbb{C}[[t, w]]) \longrightarrow G_L(\mathbb{C}), \quad \widehat{F} \mapsto \widehat{F(0, 0)}, \quad z_\lambda \mapsto z_\lambda. \quad (\text{SH40})$$

For the supported initial germ and its formal heat evolution,

$$\mathcal{H}_\sigma(\widehat{h(a + w)}) \mapsto \widehat{h(a)}, \quad \widehat{h(a)} = e \iff h(a) = 0. \quad (\text{SH41})$$

This is exactly (SH27) followed by constant-term evaluation. The multiplicity is retained as well: the order of vanishing of $h(a + w)$ is the least n with nonzero amplitude coefficient, and every coefficient is unchanged on the time-zero row. None is replaced by a different number or a different local coordinate.

After localizing at e , every value in (SH41) maps to 1_L , regardless of whether $h(a)$ vanishes. The complete connecting receiver is the pair (p, χ) in (SH3); on the germ it recovers its actual complex amplitude and its full label. The failure of χ alone to distinguish zero values is precisely its kernel congruence (SH9). For a concrete test of that defect, the two initial polynomials $z^2 - 1$ and $z^2 + 1$ have different zero locations, while their supported values, formal heat families $z^2 - 1 + 2\sigma t$ and $z^2 + 1 + 2\sigma t$, and all their localized germs have the identical full-support projection 1_L . Their amplitude projections distinguish them exactly. This constructs the lost-information object and its recovery map rather than inferring any zero-location theorem from the support quotient.

The three inverse operations therefore have explicit, different receivers: local additive inverse stays in the fibre group; local multiplicative inverse changes the identity to its specified idempotent in the local monoid; universal unital inversion receives the full lattice via q_λ . Formal heat preserves their support maps and has the exact time-zero fibre (SH25). The supported boundary (SH28) and the coefficient closure (SH33) are additional calculated objects, with their amplitude maps proved. These results do not produce an off-real zero of the original initial function, nor a conclusion for or against the Riemann hypothesis.

8 The original escaping endpoint and the arithmetic heat family

EH1. Sources and exact scope. The two supplied versions of *The flip, the Fable fibre on the Riemann sphere, and Weil positivity* have identical SHA256 values; they are retained as `inputs/flip_a.txt` and `inputs/flip_b.txt`. Their supplied attribution is Claude (Anthropic), dated 23 September 2026. They are inputs to this calculation, not primary literature certificates. The original programme calculation of the escaping orbit is retained in `material_generator_endpoint.md` and `tex/inverse_fibre_heat.tex` in the 8 September fluid research edition. All identities used from that orbit are proved again below. Its time is written τ_f here to distinguish it from the adjoined semiring zero τ . This is an explicit renaming, not a change of its time scale. The original spatial third coordinate, called t in the polynomial, is written w . No coefficient is changed.

Rodgers–Tao’s original author source defines

$$\begin{aligned} H_t(Z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du, & H_0(Z) &= \frac{1}{8} \xi \left(\frac{1}{2} + \frac{iZ}{2} \right), \\ \Phi(u) &= \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, & \xi(s) &= \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2) \zeta(s). \end{aligned} \quad (1)$$

These are equations `hoz`, `sas`, `phidef`, `htdef` in Rodgers–Tao, The de Bruijn–Newman constant is non-negative, original TeX lines 78–103. The factor $1/8$ in this definition is a nonzero scalar; the number $1/8$ in the escaping trajectory below is a time. Their different roles are resolved by an explicit map, not by matching the displayed numerals.

EH2. The original polynomial and its inverse-root chart. Let $F : \mathbb{C}_{x,y,w}^3 \rightarrow \mathbb{C}_{a,b,c}^3$ be

$$\begin{aligned} F_1 &= (1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2 y - x^3 w. \end{aligned}$$

On $x \neq 0$ set $v = 1 + xy$, $d = 2 - 3xy - x^2 w$, and $A = v^2 + v - v^3 d$, $B = 4v + 2 - 3v^2 d$. Then $F = (A/x^2, B/x, xd)$ and the coordinate Jacobian $\partial(x, v, d)/\partial(x, y, w)$ has determinant $-x^3$. The determinant of $\partial F/\partial(x, v, d)$ is x^{-3} times

$$\det \begin{pmatrix} -2A & 2v + 1 - 3v^2 d & -v^3 \\ -B & 4 - 6vd & -3v^2 \\ d & 0 & 1 \end{pmatrix} = 2.$$

Expansion reduces the last expression to $(4 - 6vd)(-2v^2 - 2v + 3v^3 d) + (2v + 1 - 3v^2 d)(4v + 2 - 6v^2 d) = 2$. Hence $\det DF = -2$ on a dense open set and therefore everywhere.

Put $P_{a,b,c}(r) = cr^3 - 2r^2 + br - 2a$. For $x \neq 0$, the exact inverse coordinates are

$$r = y + x^{-1}, \quad \alpha = x^{-1}, \quad (x, y, w) = (\alpha^{-1}, r - \alpha, 5\alpha^2 - 3r\alpha - c\alpha^3).$$

Substitution gives

$$F = (r^2 + r\alpha - cr^3, 4r + 2\alpha - 3cr^2, c).$$

Thus $F = (a, b, c)$ is equivalent to $P(r) = 0$ and $\alpha = P'(r)/2 \neq 0$. This proves both directions, including the excluded multiple-root locus. On $x = 0$ the formula is $F(0, y, w) = (w + 4y^2, y, 0)$, so the entire additional inverse point is $(0, b, a - 4b^2)$ when $c = 0$.

EH3. The escaping trajectory, with its original endpoint. For $\tau_f < 1/8$ put

$$z = \sqrt{1 - 8\tau_f} > 0, \quad \gamma(\tau_f) = (z^{-1}, -3z/2, 13z^2/2).$$

Direct substitution in the original polynomial gives

$$F(\gamma) = (-z^2/4, 0, 0) = (-1/4 + 2\tau_f, 0, 0), \quad r(\gamma) = -z/2, \quad \alpha(\gamma) = z.$$

Moreover $z' = -4/z$. Differentiating the displayed identity gives $DF(\gamma)\gamma' = (2, 0, 0)^\top$. For

$$U = 2(DF)^{-1}e_1 + 6F_3(DF)^{-1}e_2$$

this proves $\gamma' = U(\gamma)$, since $F_3(\gamma) = 0$ and DF is invertible. The trajectory has no finite affine limit because its first coordinate tends to infinity. Its target has the finite limit $(0, 0, 0)$. At that target the entire affine fibre is $(0, 0, 0)$, by the preceding inverse description: the double root $r = 0$ has $\alpha = 0$ and is excluded from the $x \neq 0$ chart. These are exact statements about this field; they require no claim concerning a separate Navier–Stokes solution.

EH4. The exact heat map and its time origin. Retain the incoming note’s arithmetic coordinate $s = 1/2 + r$ and heat coordinate $Z = -2ir$. Define

$$h = 4\tau_f - \frac{1}{2} = 2a, \quad Q_h(Z) = Z^2 - 2h.$$

On $b = c = 0$ one has the polynomial identity

$$Q_h(Z) = 2P_{a,0,0}(r) \quad (Z = -2ir, \quad h = 2a).$$

Consequently $\partial_h Q_h = -2 = -\partial_Z^2 Q_h$, with exactly the backward heat sign of (1). Along the escaping branch $Z = iz$ and $h = -z^2/2$. The original times map as

$$\tau_f = 0 \mapsto h = -1/2, \quad \tau_f \uparrow 1/8 \mapsto h \uparrow 0.$$

There is no step identifying h with the absolute arithmetic time t of (1); the formula specifies the translated and scaled time of this particular polynomial solution completely. Its zeros are $\pm i\sqrt{-2h}$ for $h < 0$, a double zero at 0 for $h = 0$, and $\pm\sqrt{2h}$ for $h > 0$. Thus its all-real threshold is exactly $h = 0$, including the endpoint. The escape reaches that real double root. It does not leave a nonreal root at the endpoint.

The reciprocal of the escaping affine coordinate has an exact relation to the meromorphic heat-root velocity. On the same $b = c = 0$ chart, $P'(r) = -4r$, so the inverse formulas give

$$\alpha = -2r = -iZ, \quad x = \frac{i}{Z}, \quad \frac{dZ}{dh} = \frac{1}{Z}, \quad x = i\frac{dZ}{dh}. \quad (2)$$

Here the derivatives are along either root branch on $Z \neq 0$: differentiating $Z^2 = 2h$ proves $2Z dZ/dh = 2$. The original root coordinate and original flow time therefore satisfy, with no change of scale left implicit,

$$h = -2r^2, \quad \frac{dr}{dh} = -\frac{1}{4r}, \quad \frac{dr}{d\tau_f} = -\frac{1}{r}, \quad \frac{dh}{d\tau_f} = 4. \quad (3)$$

The ordered-pair rational energy of the two heat roots $Z, -Z$ is

$$\mathcal{E} = \frac{2}{(2Z)^2} = \frac{1}{2Z^2} = -\frac{1}{8r^2} = -\frac{x^2}{2}. \quad (4)$$

Each equality follows from $Z = -2ir$ and $x = i/Z$. In particular, on the original escaping branch this rational function is

$$\mathcal{E}(\gamma(\tau_f)) = -\frac{1}{2z^2} = -\frac{1}{2(1-8\tau_f)} \longrightarrow -\infty. \quad (5)$$

This is its analytic continuation along imaginary heat roots $Z = iz$. On the real-root branch $h > 0$ the same rational function equals $1/(4h) > 0$ instead. Thus the reciprocal bridge retains both the factor 4 in time and the negative sign on the incoming branch. These are calculations on Q_h ; the arithmetic time t in (1) has not been identified with h .

EH5. Retaining the collision algebra and its holonomy. The family

$$\mathcal{A} = \mathbb{C}[h, Z]/(Z^2 - 2h)$$

is free of rank two over $\mathbb{C}[h]$, with basis $(1, Z)$: division by the monic polynomial gives existence and uniqueness of this basis. Its zero fibre is $\mathbb{C}[\eta]/(\eta^2)$ by $Z \mapsto \eta$. For $h \neq 0$ its two roots are exchanged by one loop around $h = 0$, since $\sqrt{2h_0}e^{i\theta} = \sqrt{2h_0}e^{i\theta/2}$ for $0 \leq \theta \leq 2\pi$. On the two labelled roots this is the matrix

$$M_{\text{loop}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad M_{\text{loop}}^* M_{\text{loop}} = I, \quad M_{\text{loop}} \neq I.$$

This computes nontrivial *unitary* monodromy around a fibre whose only root is real. It supplies an explicit example against the implication “any nonidentity holonomy forces a nonreal zero”. The map from the family to this holonomy representation is the actual root covering over $\mathbb{C}^*$, not an asserted arithmetic spectral identification.

EH6. The form through the collision. For real h , conjugation on $\mathcal{A}_h$ fixes Z . The Hermitian trace form $q_h(f, g) = \text{Tr}_{\mathcal{A}_h/\mathbb{C}}(f\bar{g})$ has matrix $\text{diag}(2, 4h)$ in $(1, Z)$. Indeed multiplication by Z has matrix $\begin{pmatrix} 0 & 2h \\ 1 & 0 \end{pmatrix}$, so its trace is zero and the trace of Z^2 is $4h$. The signature is $(2, 0)$ for $h > 0$, $(1, 1)$ for $h < 0$, and a positive line plus the nilpotent radical at $h = 0$. In the original coordinate $r = iZ/2$ with involution $r \mapsto -r$, its matrix is $\text{diag}(2, h)$: multiplying the second basis vector by $i/2$ gives precisely that congruence. On point values the reflection $s \mapsto 1 - \bar{s}$ sends Z to $\bar{Z}$, because $s = 1/2 + iZ/2$. Its induced conjugate-linear coordinate-algebra involution, given by $f^\dagger(s) = \overline{f(1 - \bar{s})}$, fixes the formal generator $Z = -2i(s - 1/2)$:

$$Z^\dagger = 2i(1/2 - s) = -2i(s - 1/2) = Z, \quad r^\dagger = -r.$$

Every sign and the degeneracy at zero therefore agree in the two presentations. No strict positivity is asserted on the nilpotent radical.

EH7. The retained split boundary of the escaping chart. Let L be the original bounded distributive support lattice and let G_L denote the original split-support semiring. Let

$$B = G_L(\mathbb{C}[z]), \quad B' = G_L(\mathbb{C}[z, z^{-1}]), \quad C = G_L(\mathbb{C}),$$

where $B \rightarrow C$ is evaluation $z \mapsto e$ (supported amplitude zero). The chart map to the trajectory is the ring homomorphism

$$\mathbb{C}[x, y, w] \longrightarrow \mathbb{C}[z, z^{-1}], \quad (x, y, w) \longmapsto (z^{-1}, -3z/2, 13z^2/2),$$

and applying G_L gives its exact support-preserving semiring map. Since B' is the localization of B at the supported polynomial z , base change along $z \mapsto e$ proves

$$B' \amalg_B C \cong C[e^{-1}] \cong L. \tag{6}$$

For clarity, the second isomorphism follows directly: $e^2 = e$ makes the image of e equal to the unit when e is invertible. Then $ea = e$ for every supported amplitude a makes all those amplitudes equal to the top support, while each lower support label stays itself. The support character χ_L witnesses that no two distinct labels are identified. These equalities also prove the universal property. The construction has thus retained a whole support spectrum at the escaping endpoint. It has not created an affine complex value for z^{-1} at $z = 0$: any attempted nonzero

ring-valued base change would send the invertible supported e to ring zero and force $1 = 0$. The exact receiver is L , with all its labels, as in (6).

EH8. The actual arithmetic coefficients, without a finite-packet replacement. For every $u \geq 0$ each summand of Φ is strictly positive, since

$$2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u} = \pi n^2 e^{5u} (2\pi n^2 e^{4u} - 3) > 0.$$

For $a = \pi e^{4u} \geq \pi$, $\sum_{n \geq 1} n^4 e^{-an^2} \leq e^{-a/2} \sum_{n \geq 1} n^4 e^{-\pi n^2/2}$. Hence $0 < \Phi(u) \leq C e^{9u} e^{-\pi e^{4u}/2}$ with a finite constant C . This bound is integrable after multiplication by any $u^N e^{Tu^2 + Ru}$, for any fixed $N, T, R \geq 0$. Uniform domination on compact subsets of $(t, Z) \in \mathbb{C}^2$ therefore proves that (1) is entire jointly in (t, Z) , and permits all the following differentiations and series integrations.

Define the exact positive moments $M_j = \int_0^\infty u^{2j} \Phi(u) du > 0$ for $j \geq 0$. The full Taylor expansion, with every time and spatial coefficient retained, is

$$H_t(Z) = \sum_{m,n \geq 0} \frac{(-1)^n M_{m+n}}{m!(2n)!} t^m Z^{2n}. \quad (7)$$

Indeed the product of the absolute exponential and cosine Taylor series is bounded on $|t| \leq T, |Z| \leq R$ by $e^{Tu^2 + Ru}$, so Tonelli and dominated convergence prove the equality. All displayed coefficients are nonzero. All coefficients of odd powers of Z are zero. The identity $\partial_t H = -\partial_Z^2 H$ follows coefficient by coefficient, including the factorials, or directly under the integral.

The actual theta reciprocal can now be compared to (2), including its first discrepancy. Fix a real arithmetic time t and define $M_j(t) = \int_0^\infty u^{2j} e^{tu^2} \Phi(u) du$. The same domination proves that each moment is finite, and strict positivity of the integrand on $u > 0$ gives $M_j(t) > 0$. Taylor expansion at the spatial origin gives

$$H_t(Z) = M_0(t) - \frac{M_1(t)}{2} Z^2 + \frac{M_2(t)}{24} Z^4 + O(Z^6). \quad (8)$$

Consequently H_Z/Z is holomorphic and nonzero at $Z = 0$, with value $-M_1(t)$. It is a holomorphic unit in a neighbourhood of zero. Division in that punctured neighbourhood yields the exact identity and Laurent expansion

$$\frac{H_{ZZ}}{H_Z} - \frac{1}{Z} = \frac{ZH_{ZZ} - H_Z}{ZH_Z} = -\frac{M_2(t)}{3M_1(t)} Z + O(Z^3), \quad (9)$$

$$\frac{H_{ZZ}}{H_Z} = \frac{1}{Z} - \frac{M_2(t)}{3M_1(t)} Z + O(Z^3). \quad (10)$$

For completeness, differentiating (8) gives $H_Z = -M_1(t)Z + M_2(t)Z^3/6 + O(Z^5)$ and $H_{ZZ} = -M_1(t) + M_2(t)Z^2/2 + O(Z^4)$. Subtracting cancels the linear numerator terms and gives $ZH_{ZZ} - H_Z = M_2(t)Z^3/3 + O(Z^5)$; its denominator is $ZH_Z = -M_1(t)Z^2 + O(Z^4)$. This proves the coefficient and its negative sign in (9).

For the quadratic, $(Q_h)_{ZZ}/(Q_h)_Z = 1/Z$ identically wherever $Z \neq 0$. The theta correction just computed is nonzero because $M_2(t)/M_1(t) > 0$. Both quotients have a simple pole of residue one at the spatial origin, but $H_t(0) = M_0(t) > 0$ for every real t , including $t = 0$. Thus this pole is at a stationary point of the actual theta function, and is not a zero of that function. The quotient is a zero velocity only after restricting to an actual simple zero. The displayed identities compute the exact discrepancy between the two meromorphic quotients without identifying their zero spaces or their absolute time coordinates.

EH9. The exact coefficient-support defect for this particular source. Represent H_0 initially as a series independent of t , with top support on every even Z coefficient in time degree zero, and bottom support elsewhere. In the coefficient-labelled semiring $(G_L(\mathbb{C}))[[t, Z]]$, lift a nonzero scalar coefficient to its top-supported value and let the heat operators retain addition's

join of supports. The minus heat operator produces the coefficients in (7). Its support is top at every $(m, 2n)$ and bottom at every $(m, 2n + 1)$.

Applying the plus heat operator to this series cancels the amplitude of every positive time degree. At $(m, 2n)$ that amplitude is

$$\frac{(-1)^n M_{m+n}}{(2n)!} \sum_{j=0}^m \frac{(-1)^j}{j!(m-j)!},$$

which is zero for $m > 0$ by $(1 - 1)^m = 0$, and equals the original $(-1)^n M_n / (2n)!$ for $m = 0$. For $m > 0$ every summand has top support, so the cancelled coefficient is the supported zero e , not absence τ . All odd coefficients remain absent. This is a calculated support memory for the actual theta source. Evaluation at $t = 0$ retains exactly the original H_0 coefficients. The general closure and its inverse-on-fixed-support interpretation are proved in the accompanying semilattice chapter.

EH10. Why the support memory alone supplies neither sign of RH. The map evaluating a globally supported function at $(0, Z_0)$ is

$$G_L(\mathcal{O}(\mathbb{C}^2)) \longrightarrow G_L(\mathbb{C}), \quad \widehat{f} \longmapsto \widehat{f(0, Z_0)}.$$

It retains every lower support label. Applied to the original source, $\widehat{H}(0, Z_0) = e$ if and only if $H_0(Z_0) = 0$, equivalently $\xi(1/2 + iZ_0/2) = 0$. Applying χ_L afterwards maps both supported zero and every supported nonzero complex value to the same top label. Thus that additional map cannot test which Z_0 is a zero. This is proved by its displayed formula, without discarding the retained support boundary.

There is also an explicit test of the amount of information in (7)'s support. Take the positive measure δ_1 on $[0, \infty)$ in the same cosine-heat integral. It gives $K_t(Z) = e^t \cos Z$, all of whose even coefficients have exactly that same nonzero sign pattern for all m, n . Every zero is real for every real t , since $\cos Z = 0$ is equivalent to $e^{2iZ} = -1$ and hence $\Im Z = 0$. Alternatively $Q_t(Z) = Z^2 - 2t$ gives the preceding nontrivial collision and monodromy at a real-root time-zero fibre. These test families are connected to the actual family by the common linear integral operator $\mu \mapsto \int e^{tu^2} \cos(Zu) d\mu$; their measures and values are stated, not substituted for $\Phi(u)du$. They show that coefficient support, local invertibility at e , and collision monodromy, without the arithmetic moment values, do not force a failure of real-rootedness at zero.

EH11. The arithmetic threshold and the exact outstanding calculation. For the original H_t , the de Bruijn–Newman theorem gives real zeros exactly for $t \geq \Lambda$. Rodgers–Tao prove $\Lambda \geq 0$ (Theorem1.1; original TeX label `main`). Platt–Trudgian's The Riemann hypothesis is true up to $3 \cdot 10^{12}$, section on the de Bruijn–Newman constant, original TeX lines135–149, proves the corollary $\Lambda \leq 0.2$ using D.H.J. Polymath, Table1. These cited results give $\text{RH} \iff \Lambda = 0$. The calculations above produce neither a nonreal zero of the actual H_0 nor a negative value of its whole Weil form. They identify the exact retained endpoint and support memory and the arithmetic evaluation map through which a proposed counterexample must pass. The assertion “RH is false because e is invertible” does not follow from these computations: the unital-localization receiver is L , and the precise local collision has a real double root. No verdict about RH is being inferred from the failure of that proposed implication.

9 Supported reciprocals in the actual zero-dynamics identities

RD1. The user's proposed contradiction and the exact source. The claim being investigated is: supported zero is invertible, therefore the heat-flow monotonicity fails at time zero, therefore RH is false. The first step must use the inverse operation actually supplied by the

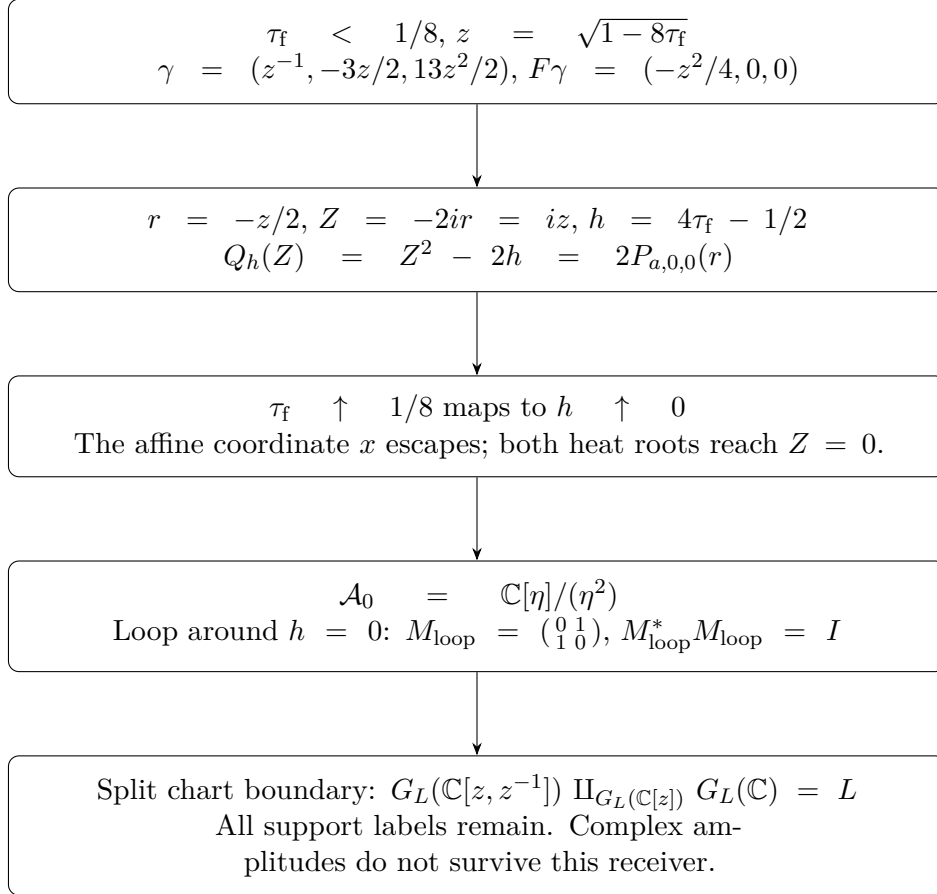


Figure 1: Exact endpoint maps of EH2–EH7. The final arrow means applying the split localization construction to the retained chart, not an identification of the dual-number algebra with L . The labels give the original coefficients and the complete time conversion. The heat coordinate convention is Rodgers–Tao’s; the polynomial orbit is the retained programme calculation rederived here.

semilattice. The second must refer to a specified monotonicity theorem. This section computes both, including a real failure of a totalized energy's monotonicity at a collision and its exact relation to heat.

The newly supplied `arXiv-1801.05914v5.tar.gz` is byte-identical to the retained Rodgers–Tao v5 original-author archive, SHA256 `cc3cd07ea418ec067bca6dce49ace0b4a2c67afe8d891fb82a1bdc5a1f63d6ed`. The actual passages read for this calculation are the definitions and proof setup, the full section *Dynamics of zeroes*, the gap propagation argument, and the full sections *Controlling the energy at time 0* and *Contradicting pair correlation*. Relevant original TeX labels are `ode`, `ident`, `tlar`, `tlar-iter`, and `picketfence`. The technical integrated energy argument is cited as part of their theorem; it is not claimed to have been reproduced here.

RD2. The original reciprocal, including supported zero. For $S = G_L(\mathbb{R})$ define

$$\widehat{a}^\dagger = \widehat{1/a} \quad (a \neq 0), \quad z_\lambda^\dagger = z_\lambda \quad (\lambda \in L).$$

This gives the multiplicative inverse in the inverse-monoid sense: $xx^\dagger x = x$, $x^\dagger xx^\dagger = x^\dagger$, and $(x^\dagger)^\dagger = x$. For nonzero amplitude these equations hold in its ordinary multiplicative group; for a support element they are the idempotence identities $z_\lambda^2 = z_\lambda$. Uniqueness follows as well. A nonzero-amplitude x forces the amplitude and top support of its inverse. If $x = z_\lambda$, the second inverse identity forces the inverse to have zero amplitude, say z_μ ; the two identities then give $\lambda \leq \mu$ and $\mu \leq \lambda$. In particular

$$e^\dagger = e, \quad ee^\dagger = e, \quad \widehat{aa}^\dagger = \widehat{1} \quad (a \neq 0).$$

This acknowledges supported zero's genuine inverse in its local multiplicative group. Replacing the middle right-hand side by $\widehat{1}$ would instead perform the unital localization $S \rightarrow L$. Both operations and their connecting maps are proved in the semilattice chapter.

RD3. Exact transport on the distinct-root locus. For a finite set of labelled real roots define $D_{jk} = x_j - x_k$ and suppose the displayed differences are nonzero. The map $a \mapsto \widehat{a}$ identifies $\mathbb{R}$ with the top additive group of S , whose identity is e . It preserves addition, multiplication, the unit and additive inverses relative to e . For every nonzero difference it also preserves the displayed reciprocal. Consequently every rational identity whose denominators are products of these differences has exactly the same lifted identity. This follows by induction on additions, multiplications and reciprocal operations in its expression. The projection $p : S \rightarrow \mathbb{R}$ is the inverse on that group. Differentiation along a top-fibre curve $\widehat{f(t)}$ means $\widehat{f'(t)}$, so the identities commute with differentiation too. Order on this fibre is the transported real order, $\widehat{a} \leq \widehat{b}$ exactly when $a \leq b$; no order is asserted between distinct lower support fibres.

For principal-value sums one first uses the exact finite identities, then takes the original amplitude limit. This defines the topology and the limit in the top fibre by the bijection p . It neither adds nor removes any summand. In particular, retaining e in the coefficient semiring changes none of the source's identities on their stated distinct-root domain.

RD4. The precise monotonicity identity, including its environment. Rodgers–Tao work under the contradiction assumption $\Lambda < 0$; their Theorem `dynam` states, for $\Lambda < t \leq 0$,

$$x'_k = 2 \sum'_{j \neq k} \frac{1}{x_k - x_j}.$$

The zeros in this range are real and simple, by the cited Csordas–Smith–Varga result. Their finite-set energy is $E^K = \sum_{k,k' \in K, k \neq k'} (x_k - x_{k'})^{-2}$, and Lemma `ident`(iii) states exactly

$$(E^K)' = \sum_{\substack{j \notin K \\ k, k' \in K, k \neq k'}} \frac{4}{(x_k - x_{k'})^2 (x_k - x_j) (x_{k'} - x_j)}$$

$$-2 \sum_{\substack{k, k' \in K \\ k \neq k'}} \left(\frac{2}{(x_k - x_{k'})^2} - \sum_{\substack{k'' \in K \\ k'' \neq k, k'}} \frac{1}{(x_{k''} - x_k)(x_{k''} - x_{k'})} \right)^2.$$

The external sum is present. A finite subsystem's energy is not asserted to decrease without controlling it. The preceding transport proof applies to every term, square, sign and cutoff here. In section 8, Proposition `tlar-iter` controls those external terms across time intervals at most $1/(100 \log^2 T)$, with an interval shrunk by $\log^3 T$ at each end. Supported-zero inversion changes none of those terms, because no denominator there is supported zero.

This reading also makes the logical direction exact. Their contradiction excludes $\Lambda < 0$, leaving both $\Lambda = 0$ and $\Lambda > 0$. The latter is entirely compatible with their theorem and is the desired RH-disproof direction. A failure of some new extension of energy monotonicity, by itself, does not establish $\Lambda > 0$; that inference requires a calculation on the original arithmetic family at its time zero, rather than on a separately extended function of colliding roots.

RD5. The full collision defect of the reciprocal substitution. The failure of cancellation at a collision is a computable object. For the map from labelled roots to coefficients,

$$\sigma(x_1, x_2) = (a, b) = (x_1 + x_2, x_1 x_2), \quad P(Z) = Z^2 - aZ + b,$$

backward heat gives the exact coefficient vector $(a', b') = (0, -2)$. Away from $x_1 = x_2$, the root velocities $x'_1 = 2/(x_1 - x_2)$, $x'_2 = 2/(x_2 - x_1)$ give precisely that vector after $D\sigma$. At (r, r) , however,

$$D\sigma_{(r,r)}(v_1, v_2) = (v_1 + v_2, r(v_1 + v_2)).$$

The cokernel is a line, explicitly received by $\ell_r(\dot{a}, \dot{b}) = \dot{b} - r\dot{a}$. Its value on the actual heat vector is -2 . This is a nonzero obstruction to any finite differentiable labelled-root lift at the collision. The semilattice generalized reciprocal sets both colliding velocities to the supported value e ; their amplitude vector is $(0, 0)$, whose image under $D\sigma$ is $(0, 0)$, leaving the same defect $(0, -2)$. Thus the full coefficient heat vector survives, while a proposed finite root vector fails by this explicitly calculated cokernel class.

The replacement object supplied by this obstruction is the ramified root space $\{(h, Z) : Z^2 = 2h\}$, where $h = Z^2/2$. Its tangent map to time vanishes at $Z = 0$, and the vector field lifting ∂_h on $Z \neq 0$ is $Z^{-1}\partial_Z$. It has a simple pole in this coordinate. Both the ramification and the pole are calculated exactly; neither is silently removed by assigning a finite reciprocal at zero.

RD6. A real loss of monotonicity after totalizing energy. For the same actual polynomial heat solution $Q_h(Z) = Z^2 - 2h$ and $h > 0$, its two roots have gap $d(h) = 2\sqrt{2h}$. Therefore its ordered-pair energy is

$$E(h) = \frac{2}{d(h)^2} = \frac{1}{4h}, \quad E'(h) = -\frac{1}{4h^2} < 0.$$

At $h = 0$ the gap is the supported zero e . If one evaluates the same expression by the generalized inverse in RD2, the assigned energy is $2e^2 = e$, of amplitude zero. Consequently the totalized function $E^\dagger(0) = 0$, $E^\dagger(h) = 1/(4h)$ for $h > 0$ is not nonincreasing on $[0, \infty)$: for every $h > 0$ it exceeds its value at zero. This proves a precise version of the user's proposed monotonicity failure.

It also determines its logical consequence. Every zero of Q_h is real for every $h \geq 0$, including the double zero at zero. Hence this failure of the extended energy's monotonicity is compatible with real-rootedness at time zero. It cannot serve as the implication from failure of that extended energy law to failure of the real-zero condition. The original energy has $\lim_{h \downarrow 0} E(h) = +\infty$; the generalized-inverse value is a different, discontinuous extension, and the two are related by the explicit reciprocal and amplitude maps.

The same distinction is visible without orders or inequalities. In the ramification coordinate $q := Z$ one has $h = q^2/2$ and $E = 1/(2q^2)$. Its Laurent germ has valuation -2 and leading coefficient $1/2$. The valued germ field $\mathbb{C}((q))$, with its valuation and residue of q^2E , retains both data; evaluation of regular functions at $q = 0$ does not extend to it as a nonzero ring homomorphism. The split special-fibre receiver of the localized chart is L as proved in (6). In the original coordinate $r = iZ/2$ the same identities are $h = -2r^2$ and $E = -1/(8r^2)$, as computed in (4). That receiver retains support; the Laurent valuation retains the pole order as additional data. This identifies exactly which boundary information is needed if an energy argument is to cross the collision.

RD7. The differential equation retained at supported time zero. At a simple zero of the actual entire function, differentiating $H_t(x(t)) = 0$ gives $x' = H_{ZZ}/H_Z$, by $\partial_t H = -H_{ZZ}$. At a double zero (t_0, x_0) where $H_{ZZ} \neq 0$, this equation in undivided form is

$$H_Z x' = H_{ZZ}.$$

Its value there has left amplitude zero for every finite velocity, and nonzero right amplitude. In the top split fibre it is the unequal pair e and $\widehat{H_{ZZ}}$. Making e a unit by localization sends both to the top support and thus solves only the support image of the equation. Applying the amplitude map before localization retains the displayed nonzero differential defect. These are exact maps between the two equations; no arithmetic sign or nonreal zero is obtained by their identification.

RD8. An actual arithmetic sign at the origin. The moments in (7) evaluate a sign without a finite-family replacement. For every real t put $M_j(t) = \int_0^\infty u^{2j} e^{tu^2} \Phi(u) du > 0$. The complete original source gives

$$H_t(0) = M_0(t) > 0, \quad H'_t(0) = 0, \quad H''_t(0) = -M_1(t) < 0,$$

and hence

$$(H'_t(0))^2 - H_t(0)H''_t(0) = M_0(t)M_1(t) > 0.$$

All derivatives and signs follow from the domination and positivity proved with (7). Thus neither spatial zero $Z = 0$ nor this curvature test at that spatial point supplies a negative witness at time $t = 0$. The spatial point and the time parameter are retained separately. No inference about the sign at other spatial points is made.

RD9. The heat shift, its time-zero fibre, and the full support lattice. This calculation tests the inverse-monoid operation while retaining the distinguished time-zero fibre. Write $c_h \in \mathbb{R}$ for the prescribed heat shift c , to distinguish it from the third target coordinate c of the original polynomial F . For every value of this retained parameter define

$$Q_t^{[c_h]}(Z) = Z^2 - 2(t - c_h) = Z^2 - 2t + 2c_h, \quad \partial_t Q_t^{[c_h]} = -\partial_Z^2 Q_t^{[c_h]} = -2. \quad (11)$$

In this test family t marks its time-zero fibre. Its use as a coordinate does not assert equality with the time of the particular arithmetic function H_t .

First keep c_h indeterminate. There is an exact isomorphism of $\mathbb{R}[c_h]$ -algebras

$$\theta : \mathbb{R}[c_h, h, Z] \longrightarrow \mathbb{R}[c_h, t, Z], \quad \theta(h) = t - c_h, \quad \theta(Z) = Z, \quad \theta^{-1}(t) = h + c_h. \quad (12)$$

The generator formulas compose to the identity and hence prove the isomorphism. They send $Z^2 - 2h$ to $Q_t^{[c_h]}$ and induce an isomorphism of the quotient root algebras. The two distinguished evaluations satisfy

$$\text{ev}_{t=0} \circ \theta = \text{ev}_{h=-c_h}, \quad \text{ev}_{t=c_h} \circ \theta = \text{ev}_{h=0}. \quad (13)$$

Thus the isomorphism retains the marked fibre by specifying its image; it does not send time zero to the collision unless $c_h = 0$. Parameter specialization to any real c_h commutes with these maps.

Here are the heat maps in their exact receiving category. Let $A = \mathbb{R}[c_h, t, Z]$. For a real increment δ set

$$\mathcal{H}_\delta f = \sum_{j \geq 0} \frac{(-\delta)^j}{j!} \partial_Z^j f, \quad \mathcal{H}_\delta Q_t^{[c_h]} = Q_{t+\delta}^{[c_h]}. \quad (14)$$

The sum is finite on each polynomial. This is an additive $\mathbb{R}[c_h, t]$ -linear map. Multiplying the two finite operator sums and collecting ∂_Z^{2n} gives the coefficient $\sum_{j=0}^n (-\delta)^j (-\epsilon)^{n-j} / (j!(n-j)!) = (-\delta - \epsilon)^n / n!$. Consequently $\mathcal{H}_\delta \mathcal{H}_\epsilon = \mathcal{H}_{\delta+\epsilon}$, and $\mathcal{H}_{-\delta}$ is its inverse under composition. The second equality in (14) follows because $\partial_Z^2 Q = 2$ and all higher even derivatives vanish. In particular $Q_t^{[c_h]}$ is exactly the formal heat image $\exp(-t\partial_Z^2)(Z^2 + 2c_h)$.

For every original bounded distributive lattice L retain the carrier

$$G_L(A) = \{(0, \lambda) : \lambda \in L\} \cup \{(f, 1_L) : f \in A\}, \quad e = (0, 1_L), \quad \tau = (0, 0_L).$$

Its addition and multiplication are respectively $(f, \lambda) + (g, \mu) = (f + g, \lambda \vee \mu)$ and $(f, \lambda)(g, \mu) = (fg, \lambda \wedge \mu)$. The lifted heat map is

$$\widetilde{\mathcal{H}}_\delta(f, \lambda) = (\mathcal{H}_\delta f, \lambda), \quad p \widetilde{\mathcal{H}}_\delta = \mathcal{H}_\delta p, \quad \chi_L \widetilde{\mathcal{H}}_\delta = \chi_L. \quad (15)$$

It is well defined because any input with support below 1_L has amplitude zero, which the additive map sends to zero. The displayed addition formula proves additivity, and scalar linearity proves $G_L(\mathbb{R}[c_h, t])$ -linearity. It fixes every z_λ , including e , and its inverse is $\widetilde{\mathcal{H}}_{-\delta}$. These are automorphisms of additive semimodules. They are not asserted to be semiring maps: for $\delta \neq 0$, $\mathcal{H}_\delta(Z)^2 = Z^2$ whereas $\mathcal{H}_\delta(Z^2) = Z^2 - 2\delta$.

In contrast, the coordinate isomorphism θ , parameter specialization, and both evaluations in (13) are ring homomorphisms. Each lifts to the unital semiring homomorphism $(f, \lambda) \mapsto (\varphi(f), \lambda)$, retaining every support label. This formula proves all corresponding squares commute. The joint map (p, χ_L) is injective, so these lifted maps retain exactly the amplitude data of the stated ring maps together with the whole original support lattice.

Now fix a real c_h . The root algebra and its time-zero fibre are

$$B_{c_h} = \mathbb{R}[t, Z]/(Z^2 - 2(t - c_h)), \quad B_{c_h}/(t) = \mathbb{R}[Z]/(Z^2 + 2c_h). \quad (16)$$

The lifted quotient $G_L(B_{c_h}) \rightarrow G_L(B_{c_h}/(t))$ identifies precisely pairs with the same support label and amplitude difference in tB_{c_h} . Indeed this follows separately from the two coordinates of its formula, and proves the asserted kernel congruence in both directions. It does not erase a support label or replace a supported zero by τ .

The real algebras in (16) can be identified completely. If $c_h < 0$, write $a = \sqrt{-2c_h} > 0$. Evaluation at a and $-a$ gives an isomorphism with $\mathbb{R} \times \mathbb{R}$: for any desired values u, v , the degree-one polynomial $(u + v)/2 + (u - v)Z/(2a)$ has those values, and the kernel is $(Z^2 - a^2)$ by division. If $c_h = 0$ the algebra is $\mathbb{R}[\eta]/(\eta^2)$, with $Z \mapsto \eta$. If $c_h > 0$, evaluation $Z \mapsto i\sqrt{2c_h}$ gives an isomorphism with $\mathbb{C}$ as a real algebra: the images of $1, Z$ form a real basis, and the relation is satisfied. These isomorphisms lift under G_L as before. They calculate the entire time-zero fibre, including its nilpotent when it is singular.

At the collision time $t = c_h$ the fibre is always $\mathbb{R}[\eta]/(\eta^2)$. This agrees with the exact ramification coordinate $q = Z$, $t = c_h + q^2/2$. The point-value receiver $G_L(\mathbb{C})$ is the same inverse monoid for every c_h , with the inverse of RD2 and $e^\dagger = e$. The collision algebra also retains its nonzero nilpotent η ; that element does not acquire a generalized inverse in the algebra. Indeed $\eta b \eta = 0 \neq \eta$ for every b , so the first inverse identity fails. After applying G_L the same amplitude calculation proves this failure for $\widehat{\eta}$. Thus the point-value inverse operation and the full singular fibre are related by their actual evaluation map $\eta \mapsto 0$; the multiplicity has not been discarded.

For real t the roots themselves are

$$\begin{cases} \pm i\sqrt{2(c_h - t)}, & t < c_h, \\ 0 \text{ with multiplicity two}, & t = c_h, \\ \pm \sqrt{2(t - c_h)}, & t > c_h. \end{cases} \quad \{t : Q_t^{[c_h]} \text{ has only real roots}\} = [c_h, \infty). \quad (17)$$

This proves that the threshold is exactly c_h and proves its behaviour at equality. At the marked time zero the roots are nonreal for $c_h > 0$, a real double root for $c_h = 0$, and distinct real roots for $c_h < 0$. The evaluation map

$$G_L(\mathbb{C}[c_h, t, Z]) \longrightarrow G_L(\mathbb{C}), \quad (f, \lambda) \longmapsto (f(c_h, t, Z), \lambda)$$

detects each root by the equality $\widehat{Q_t^{[c_h]}}(Z) = e$. Composing with χ_L sends that value, and every supported nonzero value, to 1_L . This gives the exact map that forgets the root condition; the joint receiver (p, χ_L) retains it.

Finally, on $t > c_h$ the pair gap is $d = 2\sqrt{2(t - c_h)}$ and its ordered-pair energy is

$$E_{c_h}(t) = \frac{1}{4(t - c_h)}, \quad E'_{c_h}(t) = -\frac{1}{4(t - c_h)^2}, \quad E_{c_h}^\dagger(c_h) = 0. \quad (18)$$

The last term is the amplitude of the supported value $\widehat{2}(d^\dagger)^2 = e$ at $d = e$; it is not the limit of the first term. Therefore the same failure of nonincreasing behaviour on $[c_h, \infty)$ occurs for every shift. The root-to-coefficient map still has heat vector $(0, -2)$, and at the collision $(0, 0)$ its differential has image $\{(v, 0) : v \in \mathbb{R}\}$. The cokernel functional is $\ell_0(\dot{a}, \dot{b}) = \dot{b}$, so the unresolved finite-velocity class is still exactly -2 . Time translation changes none of these local identities.

The same full lattice L , the same point-value inverse monoid, and the same genuine local inverse $e^\dagger = e$ therefore occur with every real threshold c_h . In particular, both $c_h = 0$ and $c_h > 0$ occur, so these inverse and support data alone do not choose between zero and a positive threshold, even after restricting attention to nonnegative thresholds. The singularity at the marked time zero is retained explicitly by (16); it is present precisely when $c_h = 0$. This is a test of the proposed inference from the inverse operation. The initial amplitudes $Z^2 + 2c_h$ are not the arithmetic amplitudes H_0 , whose moments and singularity question remain unchanged. No conclusion about that arithmetic singularity or RH is obtained by replacing those amplitudes with this family.

10 The source statements and the question they decide

There is no implication from an arbitrary nonidentity holonomy operator to failure of the Riemann hypothesis in the source results used here. The exact spectral implication is instead

$$\left| e^{L(\rho-1/2)} \right| \neq 1, \quad L > 0, \quad \rho \text{ an actual nontrivial zeta zero} \iff \Re \rho \neq \tfrac{1}{2}. \quad (\text{HW1})$$

The requirement that this is the actual zero representation is part of the formula, not an omitted identification. We construct its Mellin observation, full jets, primitive section, and Weil pairing below. We also construct the maps of the supported-zero boundary and the prime-orbit covers into the same arithmetic distribution.

The relevant original statements have the following exact scope. Connes–Consani, *Spectral Triples and Zeta Cycles* [1], Section 6.2, Theorem **spectralreal** (author-source lines 1378–1436), identify the spectrum on

$$\mathcal{H}(L) = (\Sigma_{eL} \mathcal{E}(\mathcal{S}_0^{\text{ev}}))^\perp \subset L^2(\mathbb{R}_+^\times / e^{L\mathbb{Z}}), \quad \mathcal{E}\phi(x) = x^{1/2} \sum_{n \geq 1} \phi(nx), \quad (\text{HW2})$$

with critical-zero ordinates satisfying $e^{i\gamma L} = 1$. Here $\mathcal{S}_0^{\text{ev}}$ imposes both $\phi(0) = 0$ and $\int_{\mathbb{R}} \phi = 0$. Their *Hochschild homology, trace map and zeta-cycles* [2], final theorem and Remark **remtauto** (lines 641–703), retain critical multiplicities in the scaling-site sheaf quotient and explicitly allow nontrivial nilpotent Jordan terms at multiple critical zeros. Their all-zero realization in

Proposition `laplacian`, evaluation map `evalmaprho`, and Corollary `rhequiv` (lines 423–476) retains all multiplicities and has scalar spectral parameter

$$\Delta(\rho) = (\rho - \tfrac{1}{2})^2 - \tfrac{1}{4}. \quad (\text{HW3})$$

The critical-cycle Hilbert space and this all-zero realization are different specified receivers, joined here by their explicit Mellin jet maps, not by discarding the noncritical coordinates.

Connes–Consani’s *Knots, primes and class field theory* [4], Proposition `mappingtorus`(iii) and Theorem `mappingtorus1`, explicitly prove nontrivial prime-orbit monodromy; their Theorem `main`(iii) identifies it with arithmetic Frobenius. Those are unconditional source results, not assertions that these monodromies refute RH. Finally, the Weil criterion concerns the full explicit formula on every admissible test. We use its original constants from [1], equations `bombieriexplicit1bis`, `bombieriexplicit2bis`, and `bombtest`, and its precise positivity statement from [3], Appendix `apppositivity`, Proposition `mainprop`. Weil’s contribution and Yoshida’s compact-test refinement are credited by that source. The bounded archimedean theorem `mainthmintro` in [3] requires support $[2^{-1/2}, 2^{1/2}]$ and vanishing Fourier values at $i/2$ and 0; it is not a universal holonomy criterion.

11 The actual arithmetic map, including every local unit

Fix

$$\begin{aligned} D &= -x\partial_x, \quad V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_+^\times) : \sup_{x>0} |x^b D^k F(x)| < \infty \ (b \in \mathbb{Z}, k \geq 0)\}, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \end{aligned} \quad (\text{HW4})$$

The Fourier transform on $\mathbb{R}$ has kernel $e^{-2\pi i x \xi}$. Poisson summation gives $\Theta\phi(x) = x^{-1} \widehat{\Theta\phi}(1/x)$ on V , so $\Theta V \subset \mathcal{B}$, with every Euler derivative included. Absolute summation for $\Re s > 1$, followed by continuation, gives

$$\mathcal{M}\Theta\phi(s) = 2\zeta(s)M_+\phi(s), \quad M_+\phi(s) = \int_0^\infty \phi(x) x^s \frac{dx}{x}, \quad \mathcal{M}DF(s) = s\mathcal{M}F(s). \quad (\text{HW5})$$

Evenness and $\phi(0) = 0$ give holomorphy of $M_+\phi$ for $\Re s > -2$; its value at 1 is zero. Thus HW5 annihilates the full multiplicity of every nontrivial zero, including all derivatives below that multiplicity.

Set $a(s) = s(s-1)$ and retain the entire completed function

$$g(s) = 2\xi(s) = a(s)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad g(1-s) = g(s), \quad g(\bar{s}) = \overline{g(s)}, \quad g(0) = g(1) = 1. \quad (\text{HW6})$$

Let Z be any finite set of its actual zeros, stable under $\iota\rho = 1 - \bar{\rho}$, with their full original orders m_ρ . For an arbitrary retained $c_h \neq 0$, define

$$\begin{aligned} h(s) &= c_h \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad A_h = \mathbb{C}[s]/(h) = \prod_{\rho \in Z} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad t_\rho = s - \rho, \\ g(\rho + t) &= u_\rho(t)t^{m_\rho}, \quad u_\rho(0) = 2\lambda_\rho \neq 0, \quad U_h = g/h, \quad u_h = [U_h] \in A_h^\times. \end{aligned} \quad (\text{HW7})$$

No factor u_ρ , c_h , or higher Taylor coefficient is replaced. The original spectral observation is

$$\mathcal{J}_h : Q \longrightarrow A_h, \quad [F] \longmapsto j_h \mathcal{M}F, \quad (j_h H)_{\rho, k} = \frac{H^{(k)}(\rho)}{k!}, \quad 0 \leq k < m_\rho. \quad (\text{HW8})$$

HW5 proves it is well defined.

Lemma 11.1 (Exact primitive full-jet section). *The map HW8 is onto. The following formulas construct a section using the actual completed zeta function and all its local units:*

$$\begin{aligned} \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, & M_+ \phi_*(s) &= \frac{1}{2}a(s)\pi^{-s/2}\Gamma(s/2), \\ R_u &= \text{rem}_h([a]^{-1}u_h^{-1}u), & S_\rho H(x) &= x^{-\rho} \int_x^\infty H(t)t^{\rho-1} dt, & S_h &= c_h^{-1} \prod_{\rho \in Z} S_\rho^{m_\rho}, \quad (\text{HW9}) \\ \text{Prim}_h(u) &= a(D)S_h\Theta(R_u(D)\phi_*), & \mathcal{M}\text{Prim}_h(u) &= aU_h R_u. \end{aligned}$$

The transform has full jet u on Z , vanishes to the full actual zero order off Z , and vanishes at $s = 0, 1$.

Proof. The Gaussian substitution $y = \pi x^2$ and the gamma recurrence give the displayed Mellin identity, including its factor $1/2$. Gaussian moments give $\phi_*(0) = 0$ and $\int \phi_* = 0$. Hence $\mathcal{M}\Theta\phi_* = g$, and $\mathcal{M}\Theta(R_u(D)\phi_*) = gR_u$. The inverse $[a]^{-1}$ exists because $0, 1 \notin Z$; u_h^{-1} is a full truncated-power-series inverse. The remainder representative has degree less than $\deg h$.

For $H \in \mathcal{B}$ with $\mathcal{M}H(\rho) = 0$, the defining integral for $S_\rho H$ can also be written $-x^{-\rho} \int_0^x H(t)t^{\rho-1} dt$. At infinity use the first formula and any sufficiently large negative power bound for H ; at zero use the second and any sufficiently large positive power bound. They prove arbitrary power bounds at both ends. Differentiation yields $(D - \rho)S_\rho H = H$, and then every Euler derivative has the same bounds. Integration by parts gives $\mathcal{M}S_\rho H = (\mathcal{M}H)/(s - \rho)$. The apparent singularity is removable by the zero assumption. Iterate this argument: division at a selected zero lowers its vanishing order by exactly one and preserves the orders at other selected zeros. Thus S_h is defined on gR_u , in any order. Mellin injectivity, obtained by Fourier injectivity on any vertical line, proves independence of that order. The result is $U_h R_u$, and application of $a(D)$ proves HW9. In A_h , $[a]u_h[R_u] = u$. At unselected zeros $h \neq 0$, so the factor $U_h = g/h$ retains the entire original zero order. Finally the factor a gives the two endpoint vanishings. $\square$

For $L \in \mathbb{R}$, retain the original unshifted dilation and its exact critical shift:

$$U_{eL}F(x) = F(e^{-L}x), \quad \mathcal{U}_L = e^{-L/2}U_{eL}, \quad T_L = M_{e^{L(s-1/2)}} : A_h \longrightarrow A_h. \quad (\text{HW10})$$

Substitution in the Mellin integral proves $\mathcal{M}\mathcal{U}_L F(s) = e^{L(s-1/2)}\mathcal{M}F(s)$. The same dilation preserves V and commutes with Θ , so it acts on Q , and

$$\mathcal{J}_h \mathcal{U}_L = T_L \mathcal{J}_h, \quad T_{\log p} = p^{-1/2} \Pi_p, \quad \Pi_p = M_{p^s}. \quad (\text{HW11})$$

Thus the relevant holonomy has an explicit arithmetic observation and every original prime factor remains present. The linear section HW9 need not commute on representatives: its exact discrepancy is

$$\mathcal{U}_L \text{Prim}_h(u) - \text{Prim}_h(T_L u) \in \ker(j_h \mathcal{M}). \quad (\text{HW12})$$

Its Mellin transform is exactly $aU_h(e^{L(s-1/2)}R_u - R_{T_L u})$. This records the defect, including its vanishing full jets. We assert no stronger identification with ΘV here.

12 Phase, modulus, and nilpotent transport

In the original local basis $1, t_\rho, \dots, t_\rho^{m_\rho-1}$ let $N_\rho = M_{t_\rho}$. All coefficients of T_L are

$$T_L|_\rho = e^{L(\rho-1/2)} \sum_{j=0}^{m_\rho-1} \frac{L^j N_\rho^j}{j!}, \quad (T_L u)_{\rho,k} = e^{L(\rho-1/2)} \sum_{j=0}^k \frac{L^{k-j}}{(k-j)!} u_{\rho,j}. \quad (\text{HW13})$$

Writing $\rho = \frac{1}{2} + \delta + i\gamma$, its scalar character is

$$e^{\delta L} e^{i\gamma L}, \quad \left| e^{L(\rho-1/2)} \right| = e^{\delta L}. \quad (\text{HW14})$$

For $L > 0$ its modulus is one exactly when $\delta = 0$; it is the identity scalar exactly when also $\gamma L \in 2\pi\mathbb{Z}$. The full $T_L|_\rho$ is the identity exactly when these two conditions hold and $m_\rho = 1$. Indeed

$$e^{LN_\rho} - 1 = N_\rho \left(L + \frac{L^2 N_\rho}{2!} + \dots \right) \quad (\text{HW15})$$

has a unit second factor for $L \neq 0$. Thus its kernel and image are precisely $\ker N_\rho$ and $\text{im } N_\rho$. In particular a multiple critical zero can have nontrivial unipotent transport while satisfying HW1 with modulus one.

Proposition 12.1 (The metric obstruction and its exact quotient). *For a fixed $L > 0$, the full packet admits a positive definite Hermitian form invariant under T_L exactly when every $\rho \in Z$ is critical and every $m_\rho = 1$. The quotient*

$$A_h \longrightarrow A_h/\mathfrak{n}_h = \mathbb{C}^Z, \quad \mathfrak{n}_h = \prod_{\rho \in Z} (t_\rho), \quad (\text{HW16})$$

always retains the scalar characters and removes precisely the nilpotent layers. Its operator has an invariant positive definite Hermitian form exactly when every selected zero is critical.

Proof. An invariant positive definite form gives the same norm to $T_L^n v$ for every integer n . For an eigenvector of eigenvalue q , this forces $|q| = 1$. If $m_\rho > 1$, apply T_L^n to 1 in that local factor: its coefficient of $t_\rho^{m_\rho-1}$ is $q^n (nL)^{m_\rho-1} / (m_\rho - 1)!$. For $|q| = 1$ this is unbounded, while every coefficient functional is bounded in a finite-dimensional positive norm. This contradicts invariance. Conversely, when all $m_\rho = 1$ and all scalar moduli are one, the coordinate counting metric is invariant. On the quotient the same proof omits the nilpotent calculation and proves the stated equivalence. $\square$

For clarity, flat circle holonomy and periodic descent can be written without an identification by name. On the logarithmic line a scalar solution $v(t) = e^{\alpha t} v(0)$ has

$$v(t + L) = e^{\alpha L} v(t). \quad (\text{HW17})$$

It defines a section of the flat line with that multiplier. The constant fibre Hermitian metric descends through this gluing exactly when $|e^{\alpha L}| = 1$. Ordinary periodic functions require the stronger condition $e^{\alpha L} = 1$. Reversing pullback orientation replaces the multiplier by its inverse, leaving both criteria unchanged. With $\alpha = i\gamma$, any L for which $\gamma L \notin 2\pi\mathbb{Z}$ gives nonidentity unitary holonomy. At $L = 0$, $T_0 = 1$ for every scalar and every jet; a loop around a parameter called zero is not this zero-time operator.

Here is the direct arithmetic content of HW2. For the Fourier character $x^{-i\gamma}$ on the circle, inverse-pullback scaling by e^b has eigenvalue $e^{i\gamma b}$. Descent requires $e^{i\gamma L} = 1$. Unfolding its Hermitian pairing with $\Sigma_{eL} \mathcal{E}\phi$ gives

$$\int_0^\infty x^{i\gamma} \mathcal{E}\phi(x) \frac{dx}{x} = \zeta\left(\frac{1}{2} + i\gamma\right) M_+ \phi\left(\frac{1}{2} + i\gamma\right). \quad (\text{HW18})$$

First prove this identity where the sum is absolutely convergent; Poisson and the two zero moments continue it to the displayed line. Periodization converges there by rapid decay at both ends, so unfolding is legitimate. With the retained ϕ_* of HW9, the second factor is nonzero on that line: $a(\frac{1}{2} + i\gamma) \neq 0$, the gamma function has no zeros, and its argument has positive real part. Thus orthogonality for every ϕ is equivalent to the critical zero condition. Fourier series on the compact circle give all its characters, proving the exact source criterion with no inference that nonidentity phase is forbidden at other lengths.

13 The original residue and the Jacobian Weil trace

Define $u^\dagger(s) = \overline{u(1 - \bar{s})}$ on the reflected packet. Retain

$$R_h(u, v) = \sum_{\rho \in Z} \text{Res}_\rho \frac{u^\dagger(s)v(s)}{g(s)} ds, \quad J_g = M_{g'(s)}. \quad (\text{HW19})$$

The coefficient matrix, with row $t_{\iota\rho}^j$ and column t_ρ^k , is exactly

$$(-1)^j [t^{m_\rho-1-j-k}] u_\rho(t)^{-1}. \quad (\text{HW20})$$

Its antidiagonal entries are $(-1)^j/(2\lambda_\rho) \neq 0$; hence the raw pairing is perfect on all original jets. Reflection sends a residue to minus its complex conjugate, so $R_h(u, v) = -\overline{R_h(v, u)}$. There is no assertion that this skew Hermitian residue is a positive definite Hilbert metric.

The logarithmic derivative gives the exact trace:

$$T_h(u, v) = R_h(u, J_g v) = \sum_{\rho \in Z} m_\rho \overline{u(\iota\rho)} v(\rho), \quad (\text{HW21})$$

$$\ker J_g = \mathfrak{n}_h, \quad \text{rad } T_h = \mathfrak{n}_h.$$

Indeed $g'/g = m_\rho/t + u'_\rho/u_\rho$, and locally $g'v = m_\rho u_\rho(0)v(\rho)t^{m_\rho-1}$ modulo t^{m_ρ} . These identities prove every assertion, including the full radical. The multiplier in HW10 obeys

$$(e^{L(s-1/2)})^\dagger = e^{-L(s-1/2)}, \quad R_h(T_L u, T_L v) = R_h(u, v), \quad T_h(T_L u, T_L v) = T_h(u, v). \quad (\text{HW22})$$

This is true even for offcritical zeros and nilpotent jets. Before the shift the two pairings are e^L -similitudes, in particular p -similitudes for the original Π_p .

At a fixed reflected zero the value block is (m_ρ) . At a nonfixed reflected pair it is

$$W_{\rho, \iota\rho} = m_\rho \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad T_L = \begin{pmatrix} q & 0 \\ 0 & 1/\bar{q} \end{pmatrix} \text{ on values, } \quad q = e^{L(\rho-1/2)}. \quad (\text{HW23})$$

Both signs are present: values $(1, 1)$ have norm $2m_\rho$ and values $(1, -1)$ have norm $-2m_\rho$. Consequently T_h is positive semidefinite exactly when every selected zero is critical, irrespective of multiplicity. For a critical multiple zero, HW22 and the degenerate positive form coexist with HW13's nontrivial unipotent factor. The perfect residue still records all the lost derivative data.

14 The whole explicit formula and an attained negative test

For $F, H \in \mathcal{B}$, set

$$\mathcal{W}(F, H) = \sum_{\rho \text{ actual}} m_\rho \overline{\mathcal{M}F(1 - \bar{\rho})} \mathcal{M}H(\rho). \quad (\text{HW24})$$

This sum and its continuity require no hypothesis on the positions inside the original critical strip. Here is a complete convergence argument. Let $H_* = \Theta\phi_*$. Its Gaussian sum satisfies $|H_*(x)| \leq Cx^4 e^{-\pi x^2}$ for $x \geq 1$, since $\sum_{n \geq 1} n^4 e^{-\pi(n^2-1)} < \infty$. The Fourier identity for the seed and Poisson give $H_*(x) = x^{-1} H_*(1/x)$. Thus

$$g(s) = \int_1^\infty H_*(x)(x^s + x^{1-s}) \frac{dx}{x}, \quad \log \max_{|s| \leq R} |g(s)| \leq C(R+1) \log(R+2). \quad (\text{HW25})$$

For the bound substitute $y = \pi x^2$, bound the resulting gamma integral by a constant times $n!$ with $n = \lceil (R+5)/2 \rceil$, and use $n! \leq n^n$. To count zeros, take $2R < r < 3R$ with a zero-free circle, factor the finitely many zeros in that disc out of g , and apply the mean-value identity

to the logarithmic modulus of the remaining nonvanishing holomorphic factor. The average of $\log |re^{i\theta} - \rho|$ is $\log r$, by its convergent logarithmic power series. Since $g(0) = 1$, this is Jensen's circle identity [5], and it yields

$$\sum_{|\rho| < r} m_\rho \log(r/|\rho|) = \frac{1}{2\pi} \int_0^{2\pi} \log |g(re^{i\theta})| d\theta. \quad (\text{HW26})$$

Every zero with $|\rho| \leq R$ contributes at least $\log 2$. Hence their total multiplicity is $O((R+1)\log(R+2))$. Repeated integration by parts in $\log x$, using HW4, bounds $\mathcal{M}F(\sigma + it)$ by every inverse power of $1 + |t|$, uniformly for $0 \leq \sigma \leq 1$, continuously in finitely many seminorms of F . Dyadic summation proves absolute convergence and continuous sesquilinearity of HW24.

Here are all signs in the explicit formula. Write

$$b_F(x) = x^{1/2}F(x), \quad b_H(x) = x^{1/2}H(x), \quad K = b_F^* * b_H, \quad b^*(x) = \overline{b(x^{-1})}, \quad (\text{HW27})$$

where convolution uses dx/x . Define

$$\begin{aligned} P(K) &= \sum_p \sum_{k \geq 1} (\log p) p^{-k/2} (K(p^k) + K(p^{-k})), \\ W_{\mathbb{R}}(K) &= (\log 4\pi + \gamma_E) K(1) \\ &\quad + \int_1^\infty (K(x) + K(x^{-1}) - 2x^{-1/2}K(1)) \frac{x^{1/2}}{x - x^{-1}} \frac{dx}{x}, \\ P_e(F, H) &= \overline{\mathcal{M}F(1)} \mathcal{M}H(0) + \overline{\mathcal{M}F(0)} \mathcal{M}H(1). \end{aligned} \quad (\text{HW28})$$

The full Guinand–Weil explicit formula in the Connes–Consani conventions cited above is

$$\mathcal{W}(F, H) = P_e(F, H) - W_{\mathbb{R}}(K) - P(K). \quad (\text{HW29})$$

For compact smooth tests this is the cited human source result: the substitution $F = x^{-1/2}b_F$ identifies its transform at every actual zero, with reflection $1 - \bar{\rho}$. In particular the endpoint product at $s = 0$ is $\overline{\mathcal{M}F(1)} \mathcal{M}H(0)$, and the product at $s = 1$ is its opposite endpoint pairing; these are not squared absolute endpoint norms.

All terms extend continuously to $\mathcal{B}$. In logarithmic coordinates $b_F(e^t)$ and all derivatives decay faster than $e^{-A|t|}$ for every fixed A ; their convolution has the same property, by splitting $|u| + |t - u| \geq |t|$. The prime sum is then bounded by convergent sums of the form $\sum_{n \geq 2} (\log n) n^{-A}$, because prime powers are integers and each has at most $\log n / \log 2$ presentations even without unique factorization. Near $x = 1$, the numerator in $W_{\mathbb{R}}$ is $O(\log x)$, and the denominator is of first order, so that integral is continuous in the needed local derivatives. At infinity the same rapid decay applies. Cutting off in $\log x$ now approximates any element of $\mathcal{B}$ in every seminorm; the continuous zero sum already proved therefore extends HW29.

Theorem 14.1 (Exact global bridge). *For all $u, v \in A_h$,*

$$\mathcal{W}(\text{Prim}_h(u), \text{Prim}_h(v)) = T_h(u, v) = R_h(u, J_g v). \quad (\text{HW30})$$

Every actual offcritical zero gives a compact smooth primitive test on which the complete Weil form is strictly negative. Conversely, a strictly negative value of that complete form on an admissible test implies at least one actual offcritical zero. Nonidentity unitary phase, higher critical jets, and the two boundary weights do not satisfy that implication by themselves.

Proof. HW9 fixes all selected jets and makes the full transform vanish to each unselected actual zero order. Therefore HW24 reduces exactly to HW21; no unselected term is dropped without its computed zero value.

For an actual nonfixed reflection pair of multiplicity m , choose Z to consist of that pair, and choose u with values $(1, -1)$ and all positive-order Taylor coefficients zero. The section $F = \text{Prim}_h(u)$ is primitive and $\mathcal{W}(F, F) = -2m$. To obtain a compact primitive test, choose $\chi \in C_c^\infty(\mathbb{R})$ equal to one near zero and put $F_n(x) = \chi(\log x/n)F(x)$. The product rule and the arbitrary power bounds in HW4 prove $F_n \rightarrow F$ in $\mathcal{B}$. Pick a nonnegative nonzero $\psi \in C_c^\infty(\mathbb{R}_+^\times)$ and $r > 1$. If $A = \mathcal{M}\psi(0) > 0$, $B = \mathcal{M}\psi(1) > 0$, the two compact functions $\psi, U_r\psi$ have endpoint matrix $\begin{pmatrix} A & A \\ B & rB \end{pmatrix}$ of determinant $AB(r-1) > 0$. Let v_0, v_1 be its inverse linear combinations giving endpoint pairs $(1, 0), (0, 1)$. Then

$$\tilde{F}_n = F_n - \mathcal{M}F_n(0)v_0 - \mathcal{M}F_n(1)v_1 \quad (\text{HW31})$$

is compact, smooth, and primitive, and converges to F in $\mathcal{B}$. By continuity its full Weil value tends to $-2m$, so is strictly negative for all sufficiently large n . This gives the test required by the criterion using the exact arithmetic map and the original local units. For the converse, if all actual zeros are critical, every summand in HW24 on the diagonal is $m_\rho |\mathcal{M}F(\rho)|^2 \geq 0$. Absolute convergence proves nonnegativity, excluding a negative test. The final assertion follows from HW13–23 and the boundary calculation below. $\square$

15 Supported zero, the endpoint representation, and inversion

The full supported theta sum and its punctured part are

$$\Theta_e\phi(x) = \sum_{n \in \mathbb{Z}} \phi(nx), \quad P\phi(x) = \sum_{n \neq 0} \phi(nx), \quad a_\phi = \phi(0), \quad b_\phi = \int_{\mathbb{R}} \phi. \quad (\text{HW32})$$

The term $n = 0$ occurs exactly once and equals a_ϕ . Poisson summation, after removing that term on both sides, gives

$$P\phi(x) = x^{-1}P\hat{\phi}(1/x) + b_\phi/x - a_\phi. \quad (\text{HW33})$$

Splitting the Mellin integral at one proves

$$\begin{aligned} \mathcal{M}P\phi(s) &= \int_1^\infty P\phi(x)x^s \frac{dx}{x} + \int_1^\infty P\hat{\phi}(x)x^{1-s} \frac{dx}{x} \\ &\quad + \frac{b_\phi}{s-1} - \frac{a_\phi}{s}. \end{aligned} \quad (\text{HW34})$$

The two integrals are entire by rapid decay. Thus supported evaluation and its Fourier dual give the original residues $-a_\phi$ at 0 and b_ϕ at 1, with their signs.

For the full bounded distributive lattice Λ , keep

$$G_\Lambda(R) = \{(r, 1_\Lambda) : r \in R\} \cup \{z_\lambda = (0, \lambda) : \lambda \neq 1_\Lambda\}, \quad \tau = z_{0_\Lambda}, \quad e = z_{1_\Lambda}, \quad (\text{HW35})$$

with componentwise addition $(r+s, \lambda \vee \mu)$ and multiplication $(rs, \lambda \wedge \mu)$. In the global fixed-sector carrier use the original space

$$\mathcal{F}_\Lambda = \mathcal{S}^{\text{ev}}(\mathbb{R}) \oplus \bigoplus_{\lambda \neq 1_\Lambda} \mathbb{C}_\lambda. \quad (\text{HW36})$$

This is the fixed-sector space, not every stalk of the full semilocal coefficient sheaf. Its exact endpoint sequence is

$$0 \longrightarrow V \longrightarrow \mathcal{F}_\Lambda \xrightarrow{\beta_\Lambda} B_\Lambda = \mathbb{C}(0)_e \oplus \mathbb{C}(1)_{e^\vee} \oplus \bigoplus_{\lambda \neq 1_\Lambda} \mathbb{C}(0)_\lambda \longrightarrow 0, \quad (\text{HW37})$$

where $\beta_\Lambda(\phi, c) = (a_\phi, b_\phi, c)$; the inclusion of V has all lower-label coordinates zero. It is onto: choose an even Schwartz function nonzero at zero, and an even Schwartz function vanishing

there with nonzero integral; their endpoint matrix is triangular and invertible, and the remaining coordinates are independent. Dilation $\phi(x) \mapsto \phi(e^{-L}x)$, fixing each c_λ , has endpoint matrix $\text{diag}(1, e^L, 1, \dots, 1)$. The same critical shift as HW10 therefore gives

$$T_L^{\text{bd}} = \text{diag}(e^{-L/2}, e^{L/2}, e^{-L/2}, \dots, e^{-L/2}). \quad (\text{HW38})$$

The top endpoint pair preserves $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Its offunit scalar moduli occur unconditionally for $L > 0$; they belong to the residues in HW34, not to the zeros in HW7. This is an actual endpoint map, so the arithmetic distinction is expressed by an exact sequence rather than an absence of any relation.

The two zero notions also have distinct operator meanings. Precomposition by multiplication by supported zero on $\mathcal{S}(\mathbb{R}) + \mathbb{C}1$ is the idempotent

$$E\phi = \phi(0)1, \quad E^2 = E, \quad \widehat{E}(\psi + c\delta_0) = \left(\int_{\mathbb{R}} \psi + c \right) \delta_0 \quad (\text{HW39})$$

on its Fourier image $\mathcal{S}(\mathbb{R}) \oplus \mathbb{C}\delta_0$. The second identity follows from $\widehat{1} = \delta_0$. Neither idempotent is an invertible transport on the whole displayed space. If an idempotent endomorphism is made invertible, it becomes the identity, and its universal linear receiver is its image: the quotient

$$X \longrightarrow X / \ker E \cong \text{im } E, \quad [x] \longmapsto Ex. \quad (\text{HW40})$$

Indeed any intertwining map into a space where $E = 1$ satisfies $f(x) = f(Ex)$, factors uniquely through that map, and $\text{im } E$ has $E = 1$. For HW39 this is the constant line, not a newly obtained nontrivial-zero eigenspace.

The same calculation on the entire lattice gives

$$G_\Lambda(R)[e^{-1}] \cong \Lambda, \quad (r, \lambda) \longmapsto \lambda. \quad (\text{HW41})$$

Proof: an invertible idempotent is one; multiplication by e sends (r, λ) to $(0, \lambda)$, so any map inverting e factors through this support projection. The lattice with unit 1_Λ realizes that factorization, proving its universal property. Every label is retained, including τ and the supported top zero.

For the actual packet define its supported semimodule

$$\mathbf{A}_\Lambda = \{(u, 1_\Lambda) : u \in A_h\} \cup \{(0, \lambda) : \lambda \neq 1_\Lambda\}. \quad (\text{HW42})$$

Addition uses support join, and the action of $(c, \lambda) \in G_\Lambda(\mathbb{C})$ uses scalar multiplication and support meet. This is closed because a nontop element has zero amplitude. The lifted arithmetic action and its precise localization are

$$\widetilde{T}_L(u, \lambda) = (T_L u, \lambda), \quad e\mathbf{A}_\Lambda = \{(0, \lambda) : \lambda \in \Lambda\} \cong \Lambda, \quad \chi\widetilde{T}_L = \chi. \quad (\text{HW43})$$

The map is additive and scalar-linear by HW10, including all lower-label zero cases; its inverse is $\widetilde{T}_{-L}$. The localization identifies amplitudes with equal support: its fibre at top is the full A_h , and every lower fibre is a singleton. It therefore has identity action on the localized labels for every T_L , whether critical, offcritical, or nonsemisimple. For an ordinary additive-group coefficient module, e already acts as zero because $e + e = e$ and additive cancellation forces $ev = 0$. Inverting it forces that coefficient module to zero. These are two exact receivers, with different categories and explicitly given maps.

$$\begin{array}{ccccc}
\mathcal{B}/\Theta V & \xrightarrow{\mathcal{J}_h} & A_h & \longrightarrow & A_h/\mathfrak{n}_h \\
\mathcal{U}_L \downarrow & & \downarrow T_L & & \downarrow \text{diag}(e^{L(\rho-1/2)}) \\
\mathcal{B}/\Theta V & \xrightarrow{\mathcal{J}_h} & A_h & \longrightarrow & A_h/\mathfrak{n}_h
\end{array}$$

Figure 2: Actual arithmetic scaling and its reduced quotient. HW8–16 prove the arrows and both commuting squares. The full-unit primitive section is HW9; the Jacobian trace descends along the last arrow by HW21. Independently, HW43 retains every lattice label and records the localization which erases amplitudes. These are exact diagrams, not identifications of every support stalk with one vector space.

16 Actual prime holonomy and its trivial-character observation

We now retain the distinct arithmetic prime-circle holonomy of [4]. Let

$$K = \widehat{\mathbb{Z}}^\times, \quad \chi : K \rightarrow G, \quad X^\chi = (Y_{\mathbb{Q}} \times G)/K, \quad (z, g) \sim (zw, \chi(w)g), \quad (\text{HW44})$$

where G is finite abelian. At prime p , put

$$I_p = \chi(\mathbb{Z}_p^\times), \quad \tilde{p}_p = 1, \quad \tilde{p}_\ell = p \ (\ell \neq p), \quad g_p = \chi(\tilde{p}), \quad \ell_p = \log p. \quad (\text{HW45})$$

The source's mapping-torus theorem gives, in the natural balanced fibre coordinate, the oriented model

$$((G/I_p) \times \mathbb{R})/((g, t) \sim (g_p^{-1}g, t + \ell_p)). \quad (\text{HW46})$$

Indeed before the finite quotient the relation is $(h, t) \sim (ph, t + \ell_p)$; reducing the compact coordinate gives $g_0 = \chi(h)^{-1}g$, which changes to $g_p^{-1}g_0$. Positive first return, expressed at $t = 0$, is multiplication by g_p . Fibre inversion turns this into the source's displayed generator convention and reverses deck characters, explaining the two orientations. At unramified primes $g_p = \text{Frob}_p$ by the source's Theorem `main`(iii).

For a finite-dimensional unitary representation $r : G \rightarrow U(V_0)$, put

$$P_{I_p} = \frac{1}{|I_p|} \sum_{i \in I_p} r(i), \quad V_p = V_0^{I_p}, \quad T_p = r(g_p)|_{V_p}. \quad (\text{HW47})$$

Finite summation proves P_{I_p} is a self-adjoint projection onto exactly the invariant subspace. Thus T_p is unitary; its eigenvalues are roots of unity. They can be nonidentity without having nonunit modulus. Sections with inverse-pullback flow have the exact model

$$v(t + \ell_p) = T_p^{-1}v(t), \quad U_p(b)v(t) = v(t - b), \quad U_p(\ell_p) = T_p. \quad (\text{HW48})$$

For $\nu \in C_c^\infty(\mathbb{R})$, Fourier expansion on each eigenline of T_p yields

$$\text{Tr} \int_{\mathbb{R}} \nu(b) U_p(b) db = \ell_p \sum_{k \in \mathbb{Z}} \text{Tr}(T_p^k) \nu(k\ell_p). \quad (\text{HW49})$$

To verify the sign, for $T_p = e^{i\theta}$ the basis is $\ell_p^{-1/2} e^{i(2\pi n - \theta)t/\ell_p}$. Integration gives $\widehat{\nu}((2\pi n - \theta)/\ell_p)$ with the minus Fourier convention. The periodic Dirac series gives $\sum_n e^{-2\pi i n b/\ell_p} = \ell_p \sum_k \delta(b - k\ell_p)$; after multiplying by $\nu(b) e^{i\theta b/\ell_p}$ it is HW49. Rapid Fourier decay proves trace-class convergence for this single circle. This does not assert trace class for the direct sum of all prime circles.

For a logarithmic test $q \in C_c^\infty(\mathbb{R})$ with transform $\mathcal{L}q(s) = \int q(t) e^{-(s-1/2)t} dt$, choose an even smooth κ which vanishes near zero and equals one when $|t| \geq \frac{1}{2} \log 2$. Apply HW49 to

$\nu_q(t) = \kappa(t)e^{-|t|/2}q(-t)$. In $\mathbb{C}[G]$, before any character projection, the resulting prime distribution is

$$\mathcal{P}_G(q) = \sum_{p,k \geq 1} (\log p) p^{-k/2} \{E_{I_p} g_p^k q(-k \log p) + E_{I_p} g_p^{-k} q(k \log p)\}, \quad E_I = \frac{1}{|I|} \sum_{i \in I} i. \quad (\text{HW50})$$

Every term and orientation follows from HW49, and the sum is finite on this compact test. Taking $\text{Tr } r$ gives the representation-valued prime trace including inertia. The cutoff has no effect at a nonzero prime iterate.

The augmentation $\epsilon : \mathbb{C}[G] \rightarrow \mathbb{C}$ has exact section $c \mapsto cE_G$, and

$$\mathcal{P}_G(q) = P_{\text{fin}}(q)E_G + (1 - E_G)\mathcal{P}_G(q), \quad \epsilon \mathcal{P}_G(q) = P_{\text{fin}}(q), \quad (\text{HW51})$$

where

$$P_{\text{fin}}(q) = \sum_{p,k \geq 1} (\log p) p^{-k/2} (q(-k \log p) + q(k \log p)). \quad (\text{HW52})$$

Proof: $\epsilon(E_I g) = 1$ and $gE_G = E_G$; the kernel of augmentation is spanned by $g - 1$, and multiplication by $1 - E_G$ projects onto it. This proves exactly how nontrivial prime holonomy reaches the original zeta distribution: its trivial character has the original HW28 prime factors, and the other character data remain in an explicitly retained augmentation ideal. More generally, for $\Re s > 1$,

$$L_r(s) = \prod_p \det(1 - p^{-s} T_p | V_p)^{-1}, \quad -\frac{L'_r}{L_r}(s) = \sum_{p,k \geq 1} (\log p) \text{Tr}(T_p^k) p^{-ks}. \quad (\text{HW53})$$

Expand $-\log(1 - z) = \sum_{k \geq 1} z^k/k$ on every eigenline; $|\text{Tr}(T_p^k)| \leq \dim V_0$ proves absolute convergence and termwise differentiation there. The trivial representation gives precisely $\zeta(s)$. No replacement of a nontrivial character by a zeta zero is made.

At the global zero every element of K stabilizes the carrier point, so HW44 gives a singleton fibre G/G . Its coefficient space is

$$W = V_0^G, \quad P_G = \frac{1}{|G|} \sum_{g \in G} r(g). \quad (\text{HW54})$$

The same holds separately at every retained fixed support label. Evaluation at zero and additive integration land in W : equivariance proves the first, and invariance of the additive Haar measure under compact units proves the second. Tensoring HW37 with W retains their weights and every label. Thus the fixed-point coefficient is $\dim W$, not $\dim V_0$; a nontrivial character has zero invariant boundary coefficient.

17 The branch loop at a double point and the attained endpoint map

For the double-root family in the heat coordinate retain

$$A_h^{\text{dbl}} = \mathbb{C}[Z]/(Z^2 - 2h). \quad (\text{HW55})$$

For $h \neq 0$, evaluation at the ordered pair $(\sqrt{2h}, -\sqrt{2h})$ identifies the algebra with $\mathbb{C}^2$. Analytic continuation around $h = 0$ interchanges the square roots: write $h = re^{i\theta}$ and continue $\sqrt{2r}e^{i\theta/2}$ from $\theta = 0$ to 2π . Consequently the loop matrix is

$$M_{\text{loop}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad M_{\text{loop}}^* M_{\text{loop}} = 1, \quad M_{\text{loop}}^2 = 1. \quad (\text{HW56})$$

It is nonidentity and unitary. At $h = 0$, the fibre is $\mathbb{C}[Z]/(Z^2)$, retaining its nonzero nilpotent Z . These are the loop of the punctured family and the nonreduced central fibre, respectively. The regular bilinear multiplication trace, in basis $(1, Z)$, is $\text{diag}(2, 4h)$; this follows by multiplying $(a + bZ)(c + dZ)$ and taking twice its constant coefficient. For real h , its real symmetric signature changes across zero while the punctured complex loop HW56 stays the same permutation. If the involution $Z \mapsto -Z$ is inserted in the first argument, that displayed diagonal is instead $\text{diag}(2, -4h)$. Both choices and their signs are explicit; neither is silently the arithmetic HW21.

The existing supported-endpoint calculation supplies an attained map, not only a formal resemblance. In the eight coordinates $(0_+, 0_-, 1_+, 1_-, -1_+, -1_-, \infty_+, \infty_-)$, let Q_J exchange $1_+ \leftrightarrow -1_-, 1_- \leftrightarrow -1_+$, and $\infty_+ \leftrightarrow \infty_-$, fixing $0_\pm$. Define

$$j(A, B) = (0, 0, A/\sqrt{2}, A/\sqrt{2}, B/\sqrt{2}, B/\sqrt{2}, 0, 0). \quad (\text{HW57})$$

Direct substitution gives the exact pullback

$$j(A, B)^* Q_J j(C, D) = \bar{A}D + \bar{B}C. \quad (\text{HW58})$$

Thus j is an isometric embedding of the hyperbolic endpoint pair of HW28. No zero value or derivative is assigned by this map: its inputs are the two actual endpoint functionals.

For completeness those inputs are attained by a precise theta test, retaining the original Fourier convention. For $q \in C_c^\infty(\mathbb{R})$, put

$$\phi_q(x) = \int_{\mathbb{R}} e^{t/2} q(t) e^{-\pi e^{2t} x^2} dt. \quad (\text{HW59})$$

This is even Schwartz, by compactness of the t -support. The Gaussian integral gives $\phi_q(0) = \mathcal{L}q(0)$ and $\int_{\mathbb{R}} \phi_q = \mathcal{L}q(1)$. The transform of a Gaussian gives $\widehat{\phi_q} = \phi_{q(-\cdot)}$. The endpoint map $q \mapsto (\mathcal{L}q(0), \mathcal{L}q(1))$ is onto: for any nonnegative compact bump of nonzero mass, that bump and its nonzero translate have endpoint column vectors differing by the factors $e^{\pm b/2}$, whose two-by-two determinant is nonzero when $b \neq 0$. Hence HW57–59 realize every endpoint direction. For a primitive test both coordinates vanish, and its image under j is zero. The whole Weil form still has the two remaining distributions in HW29.

This final diagram records the exact boundary relationship:

$$\begin{array}{ccc} C_c^\infty(\mathbb{R}) & \xrightarrow{q \mapsto \phi_q} & \mathcal{S}^{\text{ev}}(\mathbb{R}) \\ & \searrow (\mathcal{L}q(0), \mathcal{L}q(1)) & \downarrow (\text{ev}_0, f) \\ & & \mathbb{C}^2 \xrightarrow{j} \mathbb{C}^8. \end{array} \quad (\text{HW60})$$

HW34 identifies its pole residues; HW37 is its full fixed-label endpoint extension; HW9 supplies the different full-zero-jet section; HW29 joins all of them to the original explicit formula. The nontrivial loop HW56 and the endpoint embedding HW60 therefore have proved receivers, while only an actual zero character with nonunit modulus, or an attained negative value of HW29, has the RH consequence in HW1 and Theorem 14.1.

18 The retained meromorphic transport residue and its jet receiver

The original eight-state calculation also contains a different collision clock $t = h - 1/8$. It must not be replaced by the double-root clock of HW55. Retain its full local family from EHM27–46 [6]:

$$\begin{aligned} \beta(t) &= \frac{\sqrt{9 + 64t} - 3}{2}, \quad \beta(\beta + 3) = 16t, \quad \sqrt{9} = 3, \\ \mathcal{L} &= \mathbb{C}\{t\}[\epsilon, T]/(\epsilon^2 - \beta\epsilon, T^2 + 16t - (6 + 3\beta)\epsilon), \quad \mathcal{L}_0 = \mathbb{C}[T]/(T^4), \quad \epsilon = T^2/6. \end{aligned} \quad (\text{HW61})$$

The two successive monic relations prove that $1, \epsilon, T, \epsilon T$ is a free basis before specialization. Over the punctured disc the map flipping the sign only at the root $\epsilon = \beta$ is exactly

$$S_b(\epsilon) = \epsilon, \quad S_b(T) = (1 - 2\epsilon/\beta)T, \quad S_b(\epsilon T) = -\epsilon T. \quad (\text{HW62})$$

The multiplier takes values $1, -1$ on the two roots, and its square is one by $\epsilon^2 = \beta\epsilon$. It preserves the T^2 relation and is its own inverse. Since $t/\beta = (\beta + 3)/16$, its operator residue in the stated free basis is

$$\delta = tS_b|_{t=0} = -\frac{T^3}{16} \frac{d}{dT}, \quad \delta(T) = -T^3/16, \quad \ker \delta = \langle 1, T^2, T^3 \rangle, \quad \text{im } \delta = \mathbb{C}T^3, \quad \delta^2 = 0. \quad (\text{HW63})$$

Indeed its value on T is $-\frac{3}{8}\epsilon T = -T^3/16$, and its other three free-basis values are zero. The derivation sends T^4 to $-T^6/4$, so descends to the quotient. This proves its Leibniz rule and all four displayed assertions. Its nonzero coefficient rules out a regular extension of S_b in this free family. The obstruction determines the actual nonzero derivation δ , rather than a vanished transport.

It also determines the unipotent algebra action $V_b = \exp(b\delta) = 1 + b\delta$ on $\mathcal{L}_0$. The identity $\delta(f)\delta(g) = 0$, since its image is (T^3) , and the Leibniz rule prove that V_b is multiplicative; V_{-b} is its inverse. This constructed action is not misidentified as a regular specialization of the meromorphic S_b . Its regular trace pairing for $T^\# = -T$ is $4\overline{f(0)}g(0)$, because every nonconstant multiplication operator is nilpotent. Thus $Q(\delta f, g) = Q(f, \delta g) = 0$, while $\delta \neq 0$.

There is a precise linear comparison with every actual jet factor, including its scalar spectral character. For the actual local factor at ρ with $m = m_\rho \geq 2$, and any fixed $L \neq 0$, set

$$\begin{aligned} \Psi_{\rho,L} : \mathcal{L}_0 &\longrightarrow (t_\rho^{m-2}) \subset A_\rho, & \Psi_{\rho,L}(f_0 + f_1T + f_2T^2 + f_3T^3) &= f_1t_\rho^{m-2} - 16Lf_3t_\rho^{m-1}, \\ \ker \Psi_{\rho,L} &= \langle 1, T^2 \rangle, & \Psi_{\rho,L}\delta &= LN_\rho\Psi_{\rho,L}, \quad \Psi_{\rho,L}V_b = e^{-bL(\rho-1/2)}T_{bL}\Psi_{\rho,L}. \end{aligned} \quad (\text{HW64})$$

It is onto the indicated two-dimensional filtered piece, since both displayed coefficients are nonzero. Apply both sides of the middle identity to $1, T, T^2, T^3$: only T has a nonzero value, equal to Lt_ρ^{m-1} . Exponentiating, with $N_\rho^2 = 0$ on this piece, gives the final equality, retaining the scalar phase and modulus explicitly. For a simple actual zero define $\Psi_{\rho,L} = 0$; the same intertwining identity holds with zero image, and the nonzero nilpotent residue is still retained in its source. For $m = 2$, the value of $\Psi_{\rho,L}f$ is f_1 ; for $m > 2$, it is zero. Therefore HW21 evaluates its Weil pairing exactly through those values, while HW20 retains its entire raw-residue pairing with complementary jets. This gives an explicit receiver for the nilpotent defect without assigning it an unproved offcritical arithmetic character.

19 The singular endpoint in the original arithmetic heat family

EP1. The endpoint question. The requested analogy concerns a singularity at arithmetic time zero. Such a calculation can be made directly at that endpoint; it need not begin with a numerical bound on the de Bruijn–Newman constant. We retain the entire original H_t of EH1. The following statements describe every actual real zero of H_0 , with its original multiplicity and all higher coefficients. They neither assert that a multiple zero exists nor replace H_t by a freely chosen polynomial. The backward heat equation and the positivity criterion used here are those of Rodgers–Tao and Connes–Consani, cited in EH1 and HW24–31.

EP2. The full local coefficient equation. Fix an actual real zero x_0 of H_0 , let m be its order, and write

$$H_0(x_0 + \xi) = \sum_{n \geq m} a_n \xi^n, \quad a_n = \frac{\partial_Z^n H_0(x_0)}{n!}, \quad a_m \neq 0. \quad (\text{EP1})$$

The coefficients are real. Joint analyticity, established from the original theta source in EH9, and $\partial_t H = -\partial_z^2 H$ give

$$[t^j \xi^\ell] H_t(x_0 + \xi) = (-1)^j \frac{(\ell + 2j)!}{j! \ell!} a_{\ell+2j}. \quad (\text{EP2})$$

Indeed repeated differentiation gives $\partial_t^j \partial_z^\ell H = (-1)^j \partial_z^{\ell+2j} H$; evaluation at zero and division by $j! \ell!$ proves the formula. All terms with $\ell + 2j < m$ vanish. No higher coefficient has been discarded by this equality.

Define the explicit monic polynomial

$$P_n(z) = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j n!}{j! (n-2j)!} z^{n-2j}. \quad (\text{EP3})$$

In the classical Hermite convention it is $H_n^{\text{Hermite}}(z/2)$; the explicit coordinate conversion avoids any change of heat scale. The source formula is DLMF18.5.13, Chapter18, by T.H. Koornwinder, R. Wong, R. Koekoek and R.F. Swarttouw. All identities and zero properties needed below are proved here. Substitute $t = u^2$, $\xi = uz$ into the convergent local Taylor series:

$$H_{u^2}(x_0 + uz) = \sum_{n \geq m} a_n u^n P_n(z). \quad (\text{EP4})$$

For z in any fixed compact set and u sufficiently small, this is an equality of convergent analytic functions. To justify regrouping, choose a closed polydisc strictly inside a domain of analyticity of $H_t(x_0 + \xi)$. For bounded z and small u , (u^2, uz) lies in that polydisc, where its double Taylor series converges absolutely. Regrouping by $n = 2j + \ell$ is consequently legitimate.

EP3. Every leading root is real and simple. The generating function of EP3 is

$$\sum_{n \geq 0} P_n(z) \frac{w^n}{n!} = e^{zw - w^2}. \quad (\text{EP5})$$

Multiplying the two exponential series proves it coefficient by coefficient. It gives $P'_n = nP_{n-1}$, $P_{n+1} = zP_n - 2nP_{n-1}$, and $2P''_n - zP'_n + nP_n = 0$. With $w_0(z) = e^{-z^2/4}$, Taylor expansion of $w_0(z - 2w)/w_0(z)$ also proves $P_n w_0 = (-2)^n w_0^{(n)}$. Repeated integration by parts therefore gives

$$\int_{\mathbb{R}} P_n(z) R(z) e^{-z^2/4} dz = 0 \quad (\deg R < n), \quad (\text{EP6})$$

since every boundary term is a polynomial times a decaying Gaussian. If the real polynomial P_n had fewer than n real sign-changing zeros, let R be the product of their distinct linear factors. Then $\deg R < n$, while $P_n R$ has constant nonzero sign except at its finitely many zeros: all remaining real roots have even multiplicity and conjugate nonreal factors have constant positive sign on the real line. Its weighted integral is nonzero, contrary to EP6. Thus P_n has n real simple roots. Denote them by $r_1 < \dots < r_n$. This proof includes the exact weight and scale.

EP4. Actual arithmetic root branches through the endpoint. Set $c = a_{m+1}/a_m$ and $d = a_{m+2}/a_m$, retaining these actual arithmetic numbers. Dividing EP4 by $a_m u^m$ gives an analytic function of (u, z) with value $P_m(z)$ at $u = 0$. At each r_j , its z derivative is nonzero. The analytic implicit function theorem gives a unique root $z_j(u)$ with $z_j(0) = r_j$. For real u , conjugation and uniqueness make $z_j(u)$ real. To verify that these are the complete nearby zeros, choose a small disc about x_0 with no other zero of H_0 . Uniform convergence of H_t on its boundary implies $|H_t - H_0| < |H_0|$ there for small t . The straight-line homotopy has no boundary zero, so its argument-principle integral is constant and equals m . The m distinct implicit branches for small positive u exhaust that count.

Their first three terms, including the arithmetic correction, are

$$x_j(t) = x_0 + r_j\sqrt{t} + 2ct + r_j(2d - c^2)t^{3/2} + O(t^2), \quad t \downarrow 0. \quad (\text{EP7})$$

Here and below the remainders refer to this fixed actual root and its full analytic germ. For proof, substitute $z_j = r_j + A_ju + B_ju^2 + O(u^3)$ in $P_m + cuP_{m+1} + du^2P_{m+2} + O(u^3) = 0$. At a root of P_m , EP5 gives

$$P_{m+1} = -2P'_m, \quad P'_{m+1} = 0, \quad P''_m = \frac{1}{2}r_jP'_m, \quad P_{m+2} = -2r_jP'_m.$$

The successive equations give $A_j = 2c$ and $B_j = r_j(2d - c^2)$, proving EP7. In particular the order- t translation is common to every member of the colliding cluster.

For negative time put $t = -u^2$, $\xi = uz$. The leading polynomial is $i^m P_m(z/i)$, by EP2, so its roots are ir_j . For $m \geq 2$ at least two r_j are nonzero. The same implicit argument gives nonreal roots of the actual H_{-u^2} near x_0 for all sufficiently small positive u . At time zero the cluster itself consists of the real root x_0 ; at small positive time all its members are real. Thus a singular real cluster at time zero has an explicitly calculated one-sided heat transition. Its existence alone yields no nonreal zero of H_0 . This is a calculation on the actual family, not on EP3 as an independent replacement family.

EP5. The entire finite-cluster energy, including its finite part. Use the ordered-pair convention of Rodgers–Tao and keep all m roots in the cluster:

$$E_{x_0}(t) = \sum_{i \neq j} \frac{1}{(x_i(t) - x_j(t))^2}.$$

Its endpoint expansion is

$$E_{x_0}(t) = \frac{m(m-1)}{8t} + \frac{m(m-1)}{4}(c^2 - 2d) + O(t). \quad (\text{EP8})$$

This is the cluster contribution; the external interactions in RD4 remain part of the full Rodgers–Tao identity. Here is a complete derivation of its leading constant. For $P_m(z) = \prod_j (z - r_j)$, logarithmic differentiation of the product with its i th factor removed gives

$$\sum_{j \neq i} \frac{1}{r_i - r_j} = \frac{r_i}{4}, \quad \sum_{j \neq i} \frac{1}{(r_i - r_j)^2} = \frac{m-1}{6} - \frac{r_i^2}{48}. \quad (\text{EP9})$$

For the second identity, the differentiated differential equation in EP5 gives $P'''_m(r_i)/P'_m(r_i) = r_i^2/4 - (m-1)/2$. On the other hand this ratio is three times the square of the first sum minus three times the second sum. This proves EP9. The coefficient of z^{m-2} in EP3 is $-m(m-1)$ and that of z^{m-1} is zero, so $\sum_i r_i^2 = 2m(m-1)$. Summing EP9 gives $\sum_{i \neq j} (r_i - r_j)^{-2} = m(m-1)/8$. EP7 makes every gap $u(r_i - r_j)(1 + (2d - c^2)u^2) + O(u^4)$; substitution proves EP8 initially with an $O(u)$ remainder. The symmetric sum of all ordered pairs is unchanged by $u \mapsto -u$, because these two parameters describe precisely the same zero multiset of H_{u^2} . Its convergent Laurent expansion is therefore even. The remainder improves to $O(u^2) = O(t)$, as stated. For $m = 1$ the sum is empty and both displayed coefficients vanish.

For $m \geq 2$, at $t = 0$ every internal gap is supported e , so the inverse-monoid assignment sends each inverse square and their nonempty finite sum to e . For $m = 1$ the sum is empty: it is τ in the ambient semiring, or e if explicitly formed in the top additive group. For $m \geq 2$ the amplitude assigned this way is zero, whereas the Laurent germ EP8 has the positive residue $m(m-1)/8$. Both data are retained: the assignment and the Laurent pole are related by evaluation on the punctured chart, whose split special fibre is the full L . They are not identified by taking a finite limit. This calculates the exact amount of collision energy lost by that finite assignment.

EP6. The local nilpotent fibre and its Weil value. At the same actual endpoint put $\rho = 1/2 + ix_0/2$. The local algebra is

$$A_\rho = \mathbb{C}[v]/(v^m), \quad v = s - \rho = i\xi/2. \quad (\text{EP10})$$

Reflection on the point set sends ξ to $\bar{\xi}$; the induced conjugate-linear involution on its coordinate algebra fixes the generator ξ and sends v to $-v$. The regular multiplication trace is $\text{Tr}(M_f) = mf(0)$: in the ordered basis $1, v, \dots, v^{m-1}$, its matrix is triangular with diagonal $f(0)$. Consequently the exact local Weil contribution is

$$T_\rho(f, g) = m\overline{f(0)}g(0), \quad \text{rad } T_\rho = (v). \quad (\text{EP11})$$

This agrees with the actual complete-formula pullback HW21, HW24–31, including its multiplicity and both endpoint removals. It is nonnegative at this real-zero collision, and retains a nilpotent radical rather than imposing strict positivity on it. The shifted arithmetic holonomy is $e^{Lix_0/2} \exp(LM_v)$, retaining its entire nilpotent factor. It preserves EP11, since the scalar has modulus one and the nilpotent factor has constant value one.

The split lift $G_L(A_\rho)$ also retains the full nilpotent element $\hat{v}$, with $\hat{v}^m = e$. For $m > 1$ it cannot have a reflexive multiplicative inverse: projecting $\hat{v}b\hat{v} = \hat{v}$ would give $v^2p(b) = v$. Iteration gives $v = v^mp(b)^{m-1} = 0$, contradicting $v \neq 0$. Supported e itself still satisfies $e^\dagger = e$. The exact unital localization at either e or $\hat{v}$ is L : inverting $\hat{v}$ inverts its power e , and after localizing at e every top-supported amplitude becomes the top unit. This proves both universal maps and preserves all labels.

EP7. The original meromorphic holonomy residue through the pairing. HW61–64 retain the different eight-state collision clock and the original nonzero derivation $\delta = -T^3\partial_T/16$ on $\mathbb{C}[T]/(T^4)$. For an actual factor of order $m \geq 2$, their exact linear map is

$$\Psi_{\rho,L}(f) = f_1v^{m-2} - 16Lf_3v^{m-1}, \quad \Psi\delta = LM_v\Psi, \quad L \neq 0. \quad (\text{EP12})$$

For $f = T$, the image of its nonzero residue is $\Psi\delta T = Lv^{m-1}$. It is nonzero and lies in the radical of the actual Weil trace. In particular, for every packet vector g ,

$$T_h(\Psi\delta T, g) = 0, \quad R_\rho(\Psi\delta T, 1) = \frac{(-1)^{m-1}L}{u_\rho(0)} \neq 0 \quad \text{at the fixed reflected root, } L \in \mathbb{R} \setminus \{0\}. \quad (\text{EP13})$$

The second equality follows by extracting v^{-1} from $(-1)^{m-1}Lv^{m-1}/(u_\rho(v)v^m)$; it preserves the original unit $u_\rho(0) = 2\lambda_\rho$. For the first, every value of $\Psi\delta T$ is zero, and HW21 applies. Thus the holonomy defect remains detected by the perfect residue pairing, while its Weil pairing is evaluated as zero. It has not been deleted. The exact intervening map is multiplication by g' in HW19–21.

EP8. An endpoint nonreal zero and its exact persistence map. A different endpoint datum has a different consequence. For every actual nonreal zero z_0 of H_0 , choose a closed disc around it disjoint from the real axis, with zero-free boundary. Compactness gives $\min_{\partial D} |H_0| > 0$, and joint analyticity gives $\max_{\partial D} |H_t - H_0| < \min_{\partial D} |H_0|$ for all sufficiently small real $|t|$. The same homotopy/argument-principle proof as in EP4 retains the positive zero count in that disc. Thus an actual nonreal zero at time zero persists to positive times. This explains how an endpoint disproof would subsequently yield $\Lambda > 0$, without assuming that a positive-time bound has to be the starting calculation.

The endpoint defect actually evaluated in EP8 and EP13 is instead a pole of a reciprocal or an infinitesimal transport in a retained real-zero fibre. Its exact local Weil value is EP11–13. These calculations do not exhibit an actual nonreal zero or a negative complete Weil test. They do determine the pole, its finite part, the full nilpotent and support fibres, and the exact receiving

maps, which are the endpoint data required to test the proposed analogy without erasing its singularity.

EP9. The arithmetic unit receives the finite collision energy. The local unit in HW7 is determined by the same original coefficients:

$$u_\rho(v) = 16(-2i)^m \sum_{j \geq 0} a_{m+j} (-2i)^j v^j. \quad (\text{EP14})$$

Indeed $g(s) = 2\xi(s) = 16H_0(Z)$ and the local variable is $\xi_{\text{local}} = Z - x_0 = -2iv$. Dividing this exact equality by v^m gives EP14, including its factor 16. In particular

$$u_\rho(0) = 16(-2i)^m a_m, \quad \frac{u'_\rho}{u_\rho}(0) = -2ic, \quad (\log u_\rho)''(0) = 4(c^2 - 2d). \quad (\text{EP15})$$

Differentiating EP14 proves these identities; the logarithmic second derivative is independent of the chosen local logarithm. Hence

$$\text{FP}_{t=0} E_{x_0}(t) = \frac{m(m-1)}{16} (\log u_\rho)''(0). \quad (\text{EP16})$$

The raw-residue coefficient in EP13 is correspondingly $(-1)^{m-1} L / [16(-2i)^m a_m]$. In that equation Ψ is extended by zero to all other packet factors before taking the global T_h . For a check retaining the differential, set $F(\xi) = f(i\xi/2)$ and $G(\xi) = g(i\xi/2)$ for the two local jet arguments. Then

$$R_\rho(f, g) = \frac{i}{32} \text{Res}_{\xi=0} \frac{\overline{F(\xi)} G(\xi)}{H_0(x_0 + \xi)} d\xi. \quad (\text{EP17})$$

This follows from $ds = i d\xi/2$ and $2\xi(s) = 16H_0(Z)$; the use of ξ as a local variable does not change the named completed function. Inserting the actual Jacobian multiplier $g'(s) = -32iH'_0(Z)$ cancels the prefactor and gives the residue $m\overline{F(0)}G(0)$, exactly EP11. Both the energy's finite part and the two pairings are thus calculated through the same retained arithmetic unit.

EP10. The original Rodgers–Tao positive potential at supported zero. Their original section7, immediately before equation vlog, uses, on $a > 0$, the potential

$$V(a) = a^{-2} + 2a - 3 = \frac{(a-1)^2(2a+1)}{a^2} \geq 0. \quad (\text{EP18})$$

The identity follows by expansion of its numerator. Their energy uses $a = |x_k - x_j|/|\xi_k - \xi_j|$, with distinct actual roots and fixed distinct classical locations ξ_k, ξ_j ; its stated domain therefore has $a > 0$. Here we compute exactly what happens at $a = 0$.

In the original $G_L(\mathbb{R})$ let a carry top support, let $a^\dagger$ be its inverse-monoid reciprocal, and let all differences below be formed in the top additive group with identity e . Write $\eta = aa^\dagger$, an idempotent. The two expressions in EP18 have the respective extensions

$$V_{\text{expanded}} = (a^\dagger)^2 + 2a - 3, \quad V_{\text{factored}} = (a-1)^2(2a+1)(a^\dagger)^2. \quad (\text{EP19})$$

Distributing without cancelling a gives the exact defect

$$V_{\text{expanded}} - V_{\text{factored}} = (2a-3)(1-\eta). \quad (\text{EP20})$$

For proof, expand the second numerator as $2a^3 - 3a^2 + 1$, multiply by $(a^\dagger)^2$, and use $a^2(a^\dagger)^2 = \eta$ and $a^3(a^\dagger)^2 = a\eta$. No division at zero is used. On $a > 0$, $\eta = 1$ and the defect is the top additive zero e . At $a = e$, $\eta = e$ and the amplitudes are

$$p(V_{\text{expanded}}(e)) = -3, \quad p(V_{\text{factored}}(e)) = 0, \quad p(V_{\text{expanded}} - V_{\text{factored}}) = -3. \quad (\text{EP21})$$

This is a negative value of the expanded, totalized potential. Its exact origin is the idempotent complement $1 - \eta$, which vanishes on the unit locus and has amplitude 1 at the collision. Both presentations are retained. Their difference is precisely the supported-zero defect EP20. Universal unital localization at e sends both values to the same top of L and preserves every other support label, whereas the top-fibre amplitude records the difference.

For the original weighted pair the assigned expanded value is $-3/|\xi_k - \xi_j|^2$. For a nonempty colliding cluster, each such summand is negative, although the actual punctured potential has a positive pole. This extension fails the potential's original nonnegativity property at zero. It does not alter the Rodgers–Tao theorem, which uses only distinct-root values, and it is not yet a negative value of the arithmetic Weil form. The exact Weil value of the retained critical collision and its transported nilpotent is EP11–17. The two explicit receivers record where the negative assigned potential and the zero Weil contribution occur.

References

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- [7] The retained programme source calculations are *Primitive representatives of the complete arithmetic jet packets*, PL1–28 and PL37–39, in `PRIMITIVE_PACKET_LIFT.tex`; the full lattice definition and universal maps in `split_support_geometry_arithmetic_curve_v11.tex`; and the receiver calculations SZW6–15, PHW2–25, HWA27–34. All formulas used from them are rederived in this article. The private ledger records their exact source files and reading coverage. Their publicly available arithmetic receivers are SZW, equations SZW6–15 and HWA, equations HWA27–34. The present source makes no claim of publishing any previously withdrawn exploration.

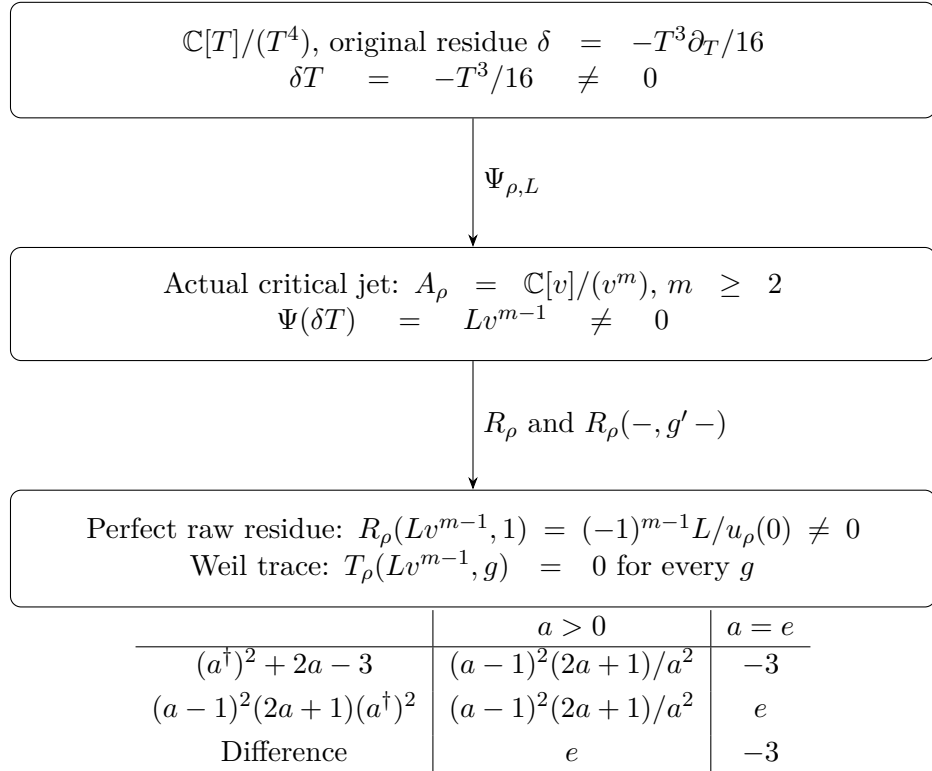


Figure 3: The endpoint defect has two explicitly computed receivers. The upper diagram retains the actual eight-state residue, its full jet image and its nonzero raw-residue value (HW61–64 and EP12–17). For a fixed reflected zero its Weil image lies in the nilpotent radical. The table computes the two extensions of Rodgers–Tao’s exact positive potential (EP18–21); entries retain top support and their differences are in the top additive group. Neither calculation identifies a negative assigned potential with a negative Weil test.